Beyond Ranks:
Inequality and the Measurement of Mobility*

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Abstract

Rank-based measures of mobility weight every positional change equally, even when the changes in income they imply are large or small. We argue for using changes in Lorenz ordinates—individuals' positions on the Lorenz curve—to measure mobility. This weights, or prices, positional changes in terms of income, so that movements implying small income changes constitute less mobility than those implying larger income changes. Lorenz mobility separates horizontal mobility (pricing positional change) from vertical mobility (changes in prices). Moreover, net Lorenz mobility requires no parent–child links to be measured. We exploit this to measure net intergenerational mobility for 81 U.S. cohorts born between 1915 and 1995. Mobility is U-shaped: the 1915 cohort ends 7 percent of aggregate income closer to the top of the income distribution than their parents, while all 55 cohorts born after 1940 end closer to the bottom—even as rank-based measures show positional mobility rising.

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1. Introduction

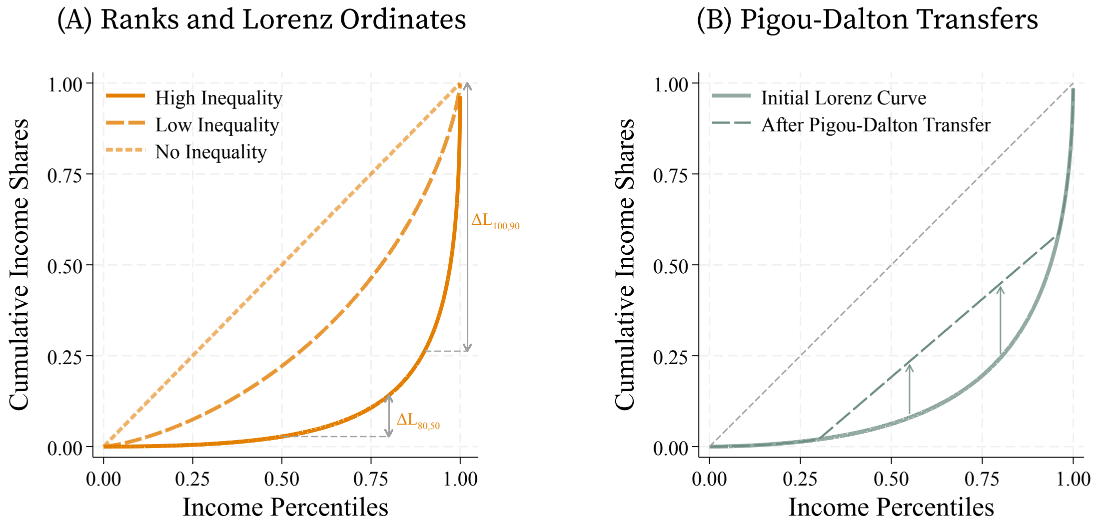
The study of mobility and inequality is intertwined, with many measures of economic mobility focusing on the *relative* position of individuals in the economy. These include intergenerational changes in income ranks (Bartholomew 1973 and Shorrocks 1978), and rank correlations (Schiller 1977, Chetty et al. 2014, or Fagereng, Mogstad, and Rønning 2021). However, purely *ordinal* measures discard *cardinal* information in income and can be driven by materially irrelevant repositioning. Large changes in ranks may have little material impact when differences in income are small, while small changes in ranks between those with drastically different incomes are meaningful. Without cardinality, relative mobility measures struggle to capture material differences.

The interaction between individuals' positions and cardinal income differences is described by the Lorenz curve—the cumulative income share at each rank. In unequal economies, income distributions compress at the bottom, where there is little difference between incomes, and spread at the top, where differences are large. Lorenz ordinates—individuals' positions on the Lorenz curve—capture this relationship: in the presence of inequality, differences in Lorenz ordinates shrink when income differences between ranks are small and spread as these differences increase as in Figure 1A. In this way, inequality determines the economic value of rank movements.

We argue this makes the *change in Lorenz ordinates* a useful measure of economic mobility. It combines ordinal differences, captured by ranks, with cardinal differences in income levels. In effect, changes in Lorenz ordinates *price* rank changes in terms of income. They tell us how the income-distance to the top decreases with upward mobility, and how it increases with downwards mobility.[1]

[1] While not driven by equity concerns, our results share properties with theories of inequality aversion (Robson 1992) and fairness (Fehr and Schmidt 1999) where the position in society is relevant (see also Lazear and Rosen 1981). However, position is not sufficient as an increase in income is valued even if ranks remain constant, underscoring the role of income for consumption capabilities (Sen 1973).

FIGURE 1. Inequality and Mobility



Notes: Panel A shows Lorenz curves for log-normal distributions with different variances, equivalently, after successive proportional Pigou-Dalton transfers. The Lorenz curve of a log-normal distribution with variance σ^2 is $L(r) = \Phi\left(\Phi^{-1}(r) - \sigma\right)$; Φ is the standard normal cumulative density. The no-inequality curve is the limit as $\sigma \longrightarrow 0$. Panel B shows a Lorenz curve before and after Pigou-Dalton transfers.

Changes in Lorenz ordinates decompose into *horizontal and vertical mobility*, making explicit the implications of inequality for mobility. Horizontal mobility comes from changes in position (or ranks) holding inequality constant: moving along the same Lorenz curve. Horizontal mobility weights positional movements proportionally to the level of income at each rank, while rank mobility weights every position equally. Thus, positional movements are meaningful insofar as they involve meaningful income changes (Figure 1A). This also implies a higher price for rank movements when they occur further up the distribution and for upward movements compared to downward movements. Vertical mobility comes from changes in inequality over time that alter the Lorenz curve. It incorporates the changes in the price schedule of rank movements.

A key implication of measuring Lorenz mobility is that reducing inequality increases mobility (by increasing Lorenz ordinates), as is the case after Pigou-Dalton transfers from richer to poorer individuals (Figure 1B).[2] In this way, our results highlight that mobility is not separable from the distribution of income, linking to extensive work interpreting mobility as social welfare (Atkinson and Bourguignon 1982). In contrast, rank-based mobility is invariant to monotone changes in the distribution, like those from progressive taxation, ignoring the effects of redistribution.

The average signed change of Lorenz ordinates—that is, *net mobility*—provides an explicit link between mobility and inequality. When aggregating over the whole population, net mobility is proportional to the change in the Gini coefficient, so that a decrease in inequality implies a net increase in mobility regardless of whether Lorenz curves intersect. As dynasties move, the Lorenz ordinate of some dynasties increases while it decreases for others, but a change towards a more equal income distribution ensures Lorenz ordinates increase on net, leading to upwards mobility. That is, a reduction in the average income-distance to the top of the distribution between generations. In this case, net mobility for the population coincides with average vertical mobility as rank changes (and therefore horizontal mobility) wash out. Furthermore, net mobility can be measured without rank linkages and therefore without panel data, a result that extends to the mobility of any sub-group of the population.

Thus, a central contribution of our paper is to extend panel independence to measuring *relative* mobility, building on existing results for *absolute* mobility measures (Ray and Genicot 2023).[3] In particular, we show net mobility does not require parent-child links to be measured; the same is true for its horizontal and vertical components. We take advantage of this to measure long-run net mobility in the U.S. for

[2] Receiving transfers brings individuals closer to those richer than them (and away from those at the bottom) even if relative positions (ranks) are preserved (Figure 1B). This reduces the income-distance to the top (and increases the distance from the bottom). It is in this sense that transfers increase mobility.

[3] Lorenz mobility complements the absolute mobility measure in Ray and Genicot (that captures income growth) by accounting for changes in the distribution of income over time and providing a measure comparable across economies with units expressed in income shares. Our approach also complements studies on top income and wealth shares (see Kopczuk 2015 for a survey). Relatedly, Audoly et al. (2025) use Lorenz ordinates to study wealth mobility as an alternative to ranks and log-wealth to emphasize mobility among top wealth holders.

81 annual cohorts born between 1915 and 1995 (Section 4). We combine repeated cross-sections from the Census, American Community Survey, and the March Current Population Survey in order to produce long-run relative mobility estimates for annual birth cohorts and to extend the analysis to racial groups and geographic regions.

Net mobility follows a pronounced U-shaped pattern. The cohort born in 1915 experiences upward mobility, being closer to the top of the income distribution than their parents by 7 percent of aggregate income, but mobility declines and becomes negative for cohorts born after 1940. By 1960, net mobility is approximately −4 percent of aggregate income, that is, Americans moved closer to the bottom of the income distribution relative to their parents. Mobility remains near that level through the 1980 cohort before partially recovering for later cohorts.

The early decline differs from historical estimates of relative mobility based on ranks, which generally rise for the cohorts born between the 1910s and 1940s (Jácome, Kuziemko, and Naidu 2025; McGee 2025). The discrepancy reflects how changes in inequality change the prices of positional moves. For instance, across the distribution, the value of a one-decile increase in position declines until 1960, except at the top where it remains high. Conversely, the loss from a one-decile fall increases over time, except at the bottom. These trends result in greater weight on downward rank changes and less on upward rank changes, yielding aggregate mobility that decreases through 1960.

Both rank and Lorenz mobility measures stabilize over similar periods for later cohorts. Rank mobility is nearly constant for the 1971–1993 birth cohorts (Chetty et al. 2014), while net Lorenz mobility is broadly flat for the cohorts born between 1960 and 1980. Nevertheless, Lorenz mobility stabilizes at a negative level, coinciding with a stabilization of how positional movements are priced. Similarly, the recovery in the latter part of the sample reflects a rise in the value of upward moves and a fall in the value of downward moves. This feature of Lorenz mobility reflects the cardinal content dispensed with by rank-based measures.

We also find sharply different paths for Black and White Americans. Net mobility for Black Americans rises initially and remains high through the 1950s before declining and converging toward the White series. Decompositions show that this advantage is driven primarily by horizontal mobility—movement in Black Americans' position in the population distribution—while the vertical component declines for both groups. The early Black–White gap in net mobility is therefore almost entirely horizontal, resulting from Black Americans' rise through the distribution, and its subsequent closing is the unwinding of that horizontal advantage rather than a change in the price of the same positional mobility.

This distinction explains why large historical *rank gains* for Black Americans (Ward 2023; Collins and Wanamaker 2022; Jácome, Kuziemko, and Naidu 2025) translate into more limited gains in aggregate-income shares—similar to results for earnings inequality in Bayer and Charles (2018). Finally, we find an early mobility advantage of people born in the South, concentrated among Black Americans.

We also show how to aggregate the changes in Lorenz ordinates across the population to obtain measures of aggregate signed mobility (Bhattacharya and Mazumder 2011; Cowell and Flachaire 2018; Ray and Genicot 2023), symmetric mobility (Fields and Ok 1996, 1999), and intergenerational correlations (Hart 1983).[4] The justification for these measures, such as the axiomatic derivation in Shorrocks (1993) for Hart's mobility measure, applies equally to mobility in Lorenz ordinates.

[4]These are commonly implemented by computing the intergenerational elasticity of income (e.g., Solon 1992; Zimmerman 1992; Dearden, Machin, and Reed 1997; Mazumder 2005; or Bolt, French, Hentall-MacCuish, and O'Dea 2024) or the Spearman correlation of income ranks (e.g., Dahl and DeLeire 2008; Olivetti and Paserman 2015; Ward 2023; or Jácome, Kuziemko, and Naidu 2025). Alternative rank-based measures are derived from transition matrices (e.g., Dardanoni 1993; D'Agostino and Dardanoni 2009).

2. Inequality and Mobility

We are interested in mobility beyond ranks, so that cardinal income changes matter *as well as* the position of individuals in the income distribution. We argue for the use of changes in Lorenz ordinates—*Lorenz mobility*—to measure mobility and proceed by establishing its properties. Most importantly, Lorenz mobility weights rank movements by the level of income at each rank, so that inequality determines the economic value of positional changes. We also give a justification for the use of Lorenz ordinates and income ranks as measures of economic *status* in Appendix A.[5]

Consider an economy with a unit mass of dynasties, $i \in [0,1]$, observed in two generations, parents and children. The income distribution of each generation is characterized by a quantile function $Q^G : [0,1] \longrightarrow \mathbb{R}$ that gives the level of income of a dynasty with rank $r \in [0,1]$ for $G \in \{P,K\}$. So, dynasty i's income in the parental generation is $y_i^P = Q^P\left(r_i^P\right)$; similarly, its income in the children's generation is $y_i^K = Q^K\left(r_i^K\right)$.[6] The Lorenz curve and average income in generation G are

$$\widetilde{L}^G(r) \equiv \int_0^r \frac{Q^G(u)}{\overline{y}^G}\,du\,, \qquad \overline{y}^G \equiv \int_0^1 Q^G(u)\,du\,, \qquad r \in [0,1]\,, \quad G \in \{P,K\}\,. \tag{1}$$

For simplicity, we refer to $L_i^G \equiv \widetilde{L}^G\left(r_i^G\right)$ as the Lorenz ordinate of dynasty i in generation $G \in \{P,K\}$. This leads to the graphical representation in Figure 1.

The key difference between Lorenz ordinates and ranks is that the former are strictly

[5]This interprets status as being *aspirational*: dynasties aspire to be ahead of others and reach those above. Status increases when a dynasty gets closer (in income) to those above and further away from those at the bottom of the distribution, given their position in the distribution. Our characterization of status via Lorenz ordinates is also linked to indexes of *aggregate satisfaction* in the context of relative deprivation (e.g., Runciman 1966; Yitzhaki 1979; Kakwani 1984), despite the two being conceptually different. Satisfaction indexes aggregate income comparisons between individuals, as Ebert and Moyes (2000) makes clear, while status captures an individual's capabilities and position in society.

[6]Equivalently, let F^G denote the cumulative distribution function of income in generation G. Then, the rank of dynasty i is $r_i^G = F^G\left(y_i^G\right)$ for $G \in \{P,K\}$.

increasing in the dynasty's income (holding the income of others constant) while the latter only increase with income if the ordering of the dynasties changes. The two measures coincide only in perfectly equal economies. So, in unequal economies, Lorenz ordinates preserve the cardinal information of income in the measurement of mobility as the Lorenz curve encodes the relative distribution of income (Atkinson 1970; Sen 1973). This makes the role of inequality in mobility explicit.

As shown in Figure 1A, a change in ranks in an unequal economy translates into a large change in Lorenz ordinates if it corresponds to a meaningful change in income. When income differences are small, as between low-income dynasties, small changes in income can generate large changes in ranks, but not meaningful changes in the consumption capabilities of those dynasty. When income differences are large, small changes in ranks can require large changes in income that are reflected in the steepness of the Lorenz curve.

Lorenz ordinates, like ranks, are bounded between 0 and 1 and have clear and interpretable units in terms of cumulative *income shares*. Specifically, Lorenz ordinates are the same under incomes $\left\{ y_i^P ,\ y_i^K \right\}$ as in an economy where average income is constant across generations, $\left\{ \bar{y}^K / \bar{y}^P \, y_i^P ,\ y_i^K \right\}$; that is, they are *scale invariant* within and between generations as in Shorrocks (1993). This renders the units in which income is measured irrelevant and separates overall income growth from mobility, so that Lorenz ordinates capture *relative* as opposed to *absolute* mobility. Consequently, an increase in the Lorenz ordinate by 0.1 for dynasty i is equivalent to the dynasty reducing their distance to the top of the distribution by 10% of aggregate income.

2.1. Inequality, redistribution, and mobility

Measuring *Lorenz mobility*—the change in Lorenz ordinates—provides a clear link between inequality, redistribution, and mobility. Concretely, mobility is the change in how much of the economy's income lies beneath the dynasty—equivalently, how much income lies above them—and movements towards equality increase upward mobility. In this way, Lorenz ordinates capture both the cardinality of income—as they increase with the dynasty's income holding the distribution constant—and the role of inequality in the economy—as they increase when the distribution becomes less unequal.

Consider two co-monotone income distributions, so that generations share the same ranks. In this case, the position of each dynasty does not change. Consequently, all mobility—between or within generations—is due to changes in inequality.[7] In contrast, rank-based measures of mobility enforce zero mobility in all cases, irrespective of the shapes of the income distributions.

More generally, the link between redistribution and mobility is made clear by contrasting the effects of *Pigou-Dalton transfers*—that redistribute income to reduce inequality—with the effects of *income swaps*—that change dynasties' positions without changing inequality.[8] Consider an arbitrary sequence of Pigou-Dalton transfers; each transfer locally reshapes the income distribution to reduce inequality. This necessarily increases Lorenz ordinates in the economy in the sense of the usual stochastic order (Atkinson 1970). The impact on mobility, however, depends on both shifts in the Lorenz curve and positional changes of dynasties after transfers, to which we return shortly in Section 2.2. Yet, compared to a sequence of income swaps that produce the same positional changes without changing inequality, Pigou-Dalton transfers increase each dynasty's mobility. Dynasties that retain or increase their rank experience greater upward mobility; conversely, those whose ranking decreases experience less

[7]Co-monotonicity only restricts the relationship between ranks, imposing no restrictions on the shape of Lorenz curves which can cross so that some dynasties experience upward mobility while others experience downward mobility. See the discussion of Axiom A5 in Appendix A.

[8]Formally, consider two dynasties h and j with $y_h < y_j$. A transfer $\delta > 0$ from j to h is a Pigou-Dalton transfer if j is not poorer than h after the transfer, $y_h + \delta \leq y_j - \delta$. These transfers maintain the relative order of the two dynasties h and j but can induce positional changes with respect to other dynasties, for instance increasing the rank of h and correspondingly decreasing the rank of a third dynasty h'.

downward mobility because the Lorenz curve after Pigou-Dalton transfers is pointwise above the original (see Figure 1B).

Proportional Pigou-Dalton transfers, which smoothly reduce inequality across the distribution, illuminate the role of pure redistribution. These transfers deliver a new income for each dynasty $y_i^{\alpha} = \alpha y_i + (1-\alpha)\mu$ for some $\alpha \in (0,1]$, and redistribute income from dynasties with above-average income to those with below-average income. They do so without changing any dynasty's position in the income distribution: preserving ranks, but raising the Lorenz ordinate of all dynasties (Figure 1B). Formally,

$$L\left(y_i^{\alpha}, Y^{\alpha}\right) = \alpha L\left(y_i, Y\right) + (1-\alpha)\, r\left(y_i, Y\right) \geq L\left(y_i, Y\right) , \tag{2}$$

where the inequality follows from the fact that $r_i \geq L_i$ for all i. This type of redistribution has the largest impact for dynasties in the middle of the distribution, even when dynasties in the extremes have the largest changes in incomes.

Pure redistribution highlights a tension between purely ordinal measures of mobility, which are insensitive to redistribution, and purely cardinal measures, which do not take the distribution of income into account. Lorenz ordinates resolve this tradeoff based on how inequality changes (something that becomes explicit when measuring average mobility, as we show in Section 3.1). They reflect cardinal changes in income affecting a single dynasty, and evaluate economy-wide changes in income in the light of changes in inequality, effectively measuring the *position* of a dynasty with the *income distance* to the bottom and the top of the distribution. This is why the Lorenz ordinate of a dynasty can increase even as their income remains constant or decreases, provided their income change is accompanied by changes in the distribution of income that "move them up," simultaneously increasing their income distance to the bottom and decreasing their distance to the top of the distribution.[9] In this way, Lorenz ordinates give the *position* of

[9] Of course, the Lorenz ordinate would decrease if the income of the dynasty decreases in isolation.

the dynasty *cardinal* content based on inequality.

Market income, disposable income, and mobility. One important implication of the link between mobility and inequality is that Lorenz mobility in market and disposable income will generally differ. This difference is driven by how the Lorenz curve of disposable income incorporates the redistribution enacted by progressive tax systems, even when taxes do not change individual positions. In contrast, *Rank-based measures of mobility are invariant to taxation* when the tax system is monotone, even if the tax system changes over time.

In general, progressivity in the children's generation increases mobility while progressivity in the parents' generation decreases it. Moreover, tax progressivity affects mobility even when it is constant over time by compressing income differences and thereby attenuating overall mobility. These results, while intuitive, are not immediate. This is because taxation does not correspond in general to an application of successive Pigou-Dalton transfers.[10] Without progressivity, as under a linear tax, taxation has no effect on the shape of the Lorenz curve or on mobility. We illustrate these ideas in the case of Pareto-distributed income in Appendix B.3.

Differentiating between market and disposable income is relevant for the interpretation of mobility. While market income is often used in the study of inequality and mobility as a measure of an individual's consumption capabilities (e.g., Piketty, Saez, and Zucman 2018), it is also understood to be an imperfect measure, not least because of the role of the tax system in redistributing resources. This makes disposable (or after-tax) income an informative measure of an individual's available resources and consumption capacity, while market income is potentially more informative about skill and earning ability.

[10] The effect of progressivity on Lorenz mobility is in stark contrast to the effect on measures of income-mobility like the intergenerational income elasticity (IGE). The IGE depends only on relative progressivity across generations so that common progressivity leaves the IGE unchanged.

2.2. Disentangling changes in position and changes in inequality

Our discussion emphasizes that Lorenz mobility reflects not only changes in a dynasty's position in society but also changes in inequality over time. We now disentangle these two forces driving mobility. We distinguish between *horizontal mobility* coming from rank changes under the same income distribution and *vertical mobility* coming from changes in inequality between the origin and the destination income distributions, while holding rank constant. In this way, Lorenz mobility can be written as

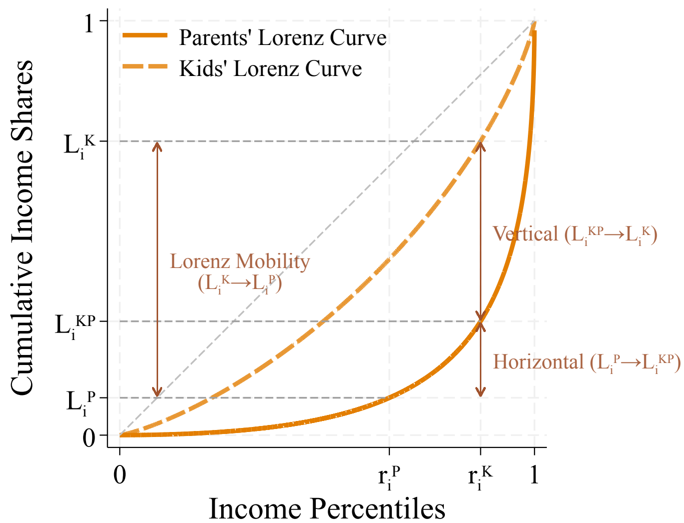
$$L_i^K - L_i^P = \overbrace{L_i^K - L_i^{KP}}^{\text{Vertical Mobility}} + \underbrace{L_i^{KP} - L_i^P}_{\text{Horizontal Mobility}}, \tag{3}$$

where $L_i^{KP} \equiv \widetilde{L}^P \left(r_i^K \right)$ is the Lorenz ordinate of generation K under P's income distribution while retaining the rank they have under K's distribution.[11] The distinction is complete in two polar cases. Mobility between co-monotone generations is entirely vertical, regardless of changes in the income distribution, while mobility between generations with the same distribution of income is entirely horizontal.

Figure 2 illustrates this. *Horizontal mobility*, the difference between L_i^{KP} and L_i^P, captures mobility coming from rank changes across generations, holding inequality constant. This corresponds to movements along generation P's Lorenz curve. *Vertical mobility*, the difference between L_i^K and L_i^{KP}, captures changes in the distribution of income across generations, holding the dynasty's rank constant. This corresponds to a (vertical) movement from generation P's Lorenz curve to K's curve. In the example, the overall decrease in inequality between generations makes vertical mobility positive for all dynasties. However, Lorenz curves can cross for general changes in the distribution

[11] It is also possible to decompose mobility prioritizing changes in inequality, moving first between Lorenz curves holding constant the parent's rank and then taking into account the change to the kid's rank. This gives $L_i^K - L_i^P = \left(L_i^{PK} - L_i^P\right) + \left(L_i^K - L_i^{PK}\right)$.

FIGURE 2. Horizontal and Vertical Mobility



Notes: The figure illustrates horizontal and vertical mobility for a dynasty moving from rank r_i^P in the parents' Lorenz curve (dark solid line) to rank r_i^K in the kids' curve (light dashed line). Horizontal mobility corresponds to the movement from L_i^P to the Lorenz ordinate of an individual with rank r_i^K in the parents' Lorenz curve, L_i^{KP}. Vertical mobility corresponds to the movement from L_i^{KP} to the ordinate L_i^K.

of income, so that some dynasties experience positive vertical mobility while others experience negative vertical mobility.

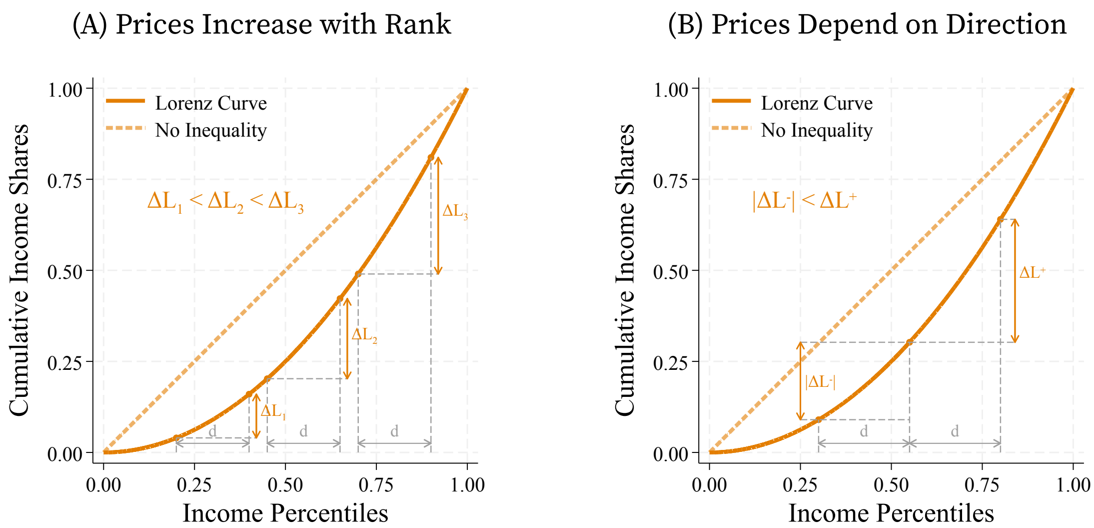
2.3. Pricing rank movements

Lorenz ordinates assign an economic value to changes in rank. This is more easily seen when we hold the income distribution fixed across generations, so that all mobility is horizontal and does not reflect changes in inequality. For a dynasty i moving from rank r_i^P to rank r_i^K, rank (Bartholomew 1973) and Lorenz mobility are, respectively,

$$\text{Rank Mobility:} \qquad m_i^R = r_i^K - r_i^P = \int_{r_i^P}^{r_i^K} 1\, du\,, \tag{4}$$

$$\text{Lorenz Mobility:} \qquad m_i^L = \widetilde{L}(r_i^K) - \widetilde{L}(r_i^P) = \int_{r_i^P}^{r_i^K} \underbrace{\frac{Q(u)}{\overline{y}}}_{\text{Price of rank movement } p_L(u)} du\,. \tag{5}$$

FIGURE 3. The Price of Rank Mobility



Notes: Panel A shows the Lorenz value of rank gains of magnitude d at different points of the income distribution. Panel B compares equal upward and downward positional movements starting from the same position r. Along the 45-degree equality line, every rank movement is priced equally.

Thus, ranks price every interval equally, while Lorenz mobility uses the relative-income schedule $p_L(u) \equiv Q(u)/\overline{y}$ to price the rank movement of the dynasty. That is, Lorenz mobility weights positional movements by the *level of income* at each rank.

The price schedule p_L is weakly increasing because income increases with rank. Consequently, the same rank movement has a higher price when it occurs further up the distribution as we show in Figure 3A. The same logic implies that directional moves are priced asymmetrically, so that an increase in rank is priced more highly than a decrease starting from the same position as in Figure 3B. Only under perfect equality does Lorenz mobility coincides with rank mobility and $p_L(u) = 1$. In this precise sense, inequality determines the economic value of a rank movement.

LEMMA 1 (**Pricing Rank Movements**). *Define the prices of upward and downward horizontal Lorenz mobility starting from a rank r and moving d* > 0 *ranks as*

$$\pi_L^{(+)}(r,d) \equiv \widetilde{L}(r+d) - \widetilde{L}(r)\,, \qquad \pi_L^{(-)}(r,d) \equiv \widetilde{L}(r) - \widetilde{L}(r-d)\,. \tag{6}$$

Then:

(a) *The prices of upward and downward Lorenz mobility are increasing in the initial rank.*

$$\textit{If} \quad s > r \quad \longrightarrow \quad \pi_L^{(+)}(s,d) \ \geq \ \pi_L^{(+)}(r,d) \quad \textit{and} \quad \pi_L^{(-)}(s,d) \ \geq \ \pi_L^{(-)}(r,d)\ . \tag{7}$$

(b) *The price of upward mobility is greater than the price of downward mobility*

$$\forall\, r \in \left[d, 1-d\right] \quad \pi_L^{(+)}(r,d) \ \geq \ \pi_L^{(-)}(r,d)\ . \tag{8}$$

PROOF. The results follow from the convexity and monotonicity of the Lorenz curve. The price schedule $p_L(u) = Q(u)/\mu$ is weakly increasing in rank. Equation (5) gives

$$\pi_L^{(+)}(r,d) \ = \ \int_0^d p_L(r+u)du\ , \qquad\qquad \pi_L^{(-)}(r,d) \ = \ \int_0^d p_L(r-d+u)du\ . \tag{9}$$

Moving the initial rank from r to $s > r$ shifts the integrands in both expressions to weakly higher ranks, proving the first result. Moreover, $p_L(r+u) \geq p_L(r-d+u)$ for every $u \in [0,d]$, proving the second. Strict convexity makes p_L strictly increasing over the relevant intervals, while under perfect equality $p_L(u) = 1$ everywhere. □

Rank mobility records movements of d in every case; Lorenz mobility emphasizes that their economic value depends on their origin and direction. When inequality changes over time, vertical mobility adds to the pricing of horizontal mobility by incorporating the change in the price schedule:

$$L_i^K - L_i^P \ = \ \overbrace{\int_0^{r_i^K} p_L^K(u) - p_L^P(u)\, du}^{\text{Vertical Mobility: } L_i^K - L_i^{KP}} \ + \ \underbrace{\int_{r_i^P}^{r_i^K} p_L^P(u)\, du}_{\text{Horizontal Mobility: } L_i^{KP} - L_i^P}\ . \tag{10}$$

3. Aggregating mobility

We consider two main measures of dynastic mobility, each defined for a dynasty and aggregated over the population. These are signed and symmetric mobility, measured respectively by the difference and absolute difference between the Lorenz ordinate of the dynasty in the kid's and parent's generations:

$$\text{Signed mobility:} \qquad m\left(L^P, L^K\right) = L^K - L^P\,; \tag{11}$$

$$\text{Symmetric mobility:} \qquad m\left(L^P, L^K\right) = \left|L^K - L^P\right|\,. \tag{12}$$

The first measure captures cases in which society values the *direction* of mobility. The second measure accounts for all mobility, regardless of direction, similar to Fields and Ok's 1996 measure for mobility in income levels. We provide conditions on the properties of mobility that lead to these measures in Appendix B.

3.1. Average mobility, inequality, and panel independence of mobility measures

Average mobility, $\overline{m} \equiv \int m_i\, di$, is linked directly to changes in inequality. This is more transparent when aggregating signed mobility (as in 11) to obtain *net mobility*. When aggregating for the whole population, net mobility is given by the change in the Gini coefficient between generations and captures exactly changes in inequality over time.

Lemma 2 (**Net Mobility**). *Under mobility (11), net mobility for a group $\mathcal{I}$ with measure $M_{\mathcal{I}}$ satisfies:*

$$\overline{m}^s_{\mathcal{I}} \equiv \frac{1}{M_{\mathcal{I}}} \int_{i\in\mathcal{I}} m^s_i\, di = \frac{1}{M_{\mathcal{I}}}\left(\int_{i\in\mathcal{I}} L^K_i\, di - \int_{i\in\mathcal{I}} L^P_i\, di\right)\,. \tag{13}$$

Net mobility for the whole population corresponds to the change in the Gini coefficient:

$$\overline{m}^s \equiv \int_0^1 m^s_i\, di = \frac{1}{2}\left(\mathcal{G}^P - \mathcal{G}^K\right)\,, \tag{14}$$

where $\mathcal{G}^G = 1 - 2\int_0^1 \widetilde{L}^G(r)\, dr$ is the Gini coefficient.

Proof. Equations (13) and (14) follow from the definition of net mobility and of $\mathcal{G}$. □

When inequality decreases, so $\mathcal{G}^K < \mathcal{G}^P$, population net mobility is positive. A more equal income distribution raises Lorenz ordinates for some dynasties and lowers them for others, but they increase on net and lead to net upwards mobility, regardless of population reshuffling. For instance, proportional Pigou-Dalton transfers lower the Gini coefficient and increase mobility because they raise the Lorenz curve of all dynasties (see equation 2 and Figure 1A).[12] This result holds in general, even under arbitrary Pigou-Dalton transfers when dynasties are subject to both vertical and horizontal mobility as in (3). This is because net mobility for the population as a whole is entirely captured by changes in the Gini coefficient or vertical mobility (horizontal mobility, like rank mobility, is zero-sum as we show in 16).[13]

Moreover, the characterization of net mobility in terms of changes in the Lorenz curve makes it *panel independent*. It can therefore be measured from repeated cross-sectional data or datasets without inter-generational links (as in Ray and Genicot 2023). In fact, the vertical component of mobility is always panel independent (including for symmetric mobility) as is the horizontal component of net mobility (the joint distribution of ranks is only needed for the concentration of horizontal mobility).

LEMMA 3 (**Panel Independence of Mobility Measures**). *The measurement of vertical mobility is panel independent as it does not depend on the joint distribution of ranks across generations. The same applies to the measurement of net mobility and signed horizontal mobility for the population or any of its sub-groups.*

PROOF. For any measure of mobility $\widetilde{m}\left(L_i^K, L_i^P\right)$, vertical mobility corresponds to $\widetilde{m_i}^V = \widetilde{m}\left(\widetilde{L}^K(r_i^K), \widetilde{L}^P(r_i^K)\right)$ as per (3). The measurement of $\widetilde{m_i}^V$ involves each generation's Lorenz curve ($\widetilde{L}^K$ and $\widetilde{L}^P$) evaluated at the children's generation ranks, which are uniformly distributed. The joint distribution of ranks is never used.

For the measurement of net mobility, Lemma 2 establishes that it corresponds to the change in the average Lorenz ordinate of each generation (or the Gini coefficient for population mobility), each an attribute of the corresponding generation's marginal

[12] This result generalizes when aggregating *status mobility* as defined in Section A. Status depends on weighted Lorenz ordinates and ranks so that net mobility corresponds to Weymark's 1981 Generalized-Gini coefficient, $\widetilde{\mathcal{G}} \equiv 1 - 2\int_0^1 \Omega(r)\,\widetilde{L}(r)\;dr$ for positive weights $\Omega(\cdot)$. See Lemma A3 in Appendix B.

[13] Similar to Ray and Genicot (2023), net mobility eliminates exchange mobility. It "sets aside the "pure movement" component of mobility" [pg. 3044] and in our case focuses on the overall change in the position of individuals measured by their income-distance to the top and bottom of the distribution.

distribution. Finally, net horizontal mobility is identically zero for the whole population and corresponds to

$$\overline{m}^{s,H} = \frac{1}{M_{\mathcal{I}}} \left(\int_{i \in \mathcal{I}} \widetilde{L}^P \left(r_i^K \right) \, di - \int_{i \in \mathcal{I}} \widetilde{L}^P \left(r_i^P \right) \, di \right) \tag{15}$$

for a group $\mathcal{I}$, which also does not make use of the joint distribution of ranks. $\square$

These properties of signed mobility greatly facilitate the measurement of aggregate mobility, as most data sources on income lack the longitudinal dimension required to follow dynasties (or individuals) over time. We return to this in Section 4 where we compute mobility for the US over 80 years.

Finally, average *symmetric mobility* is also directly linked to the change in inequality, as captured by the Gini coefficient. We show in Lemma A2 that symmetric mobility is equal to the absolute value of net mobility, adjusted by either downward or upward Lorenz mobility depending on whether overall inequality has increased or decreased.[14] Average symmetric mobility therefore reflects gains and losses in Lorenz ordinates arising from dynasties' positional changes, as well as gains or losses arising from changes in the shape of the Lorenz curve that are not summarized by the Gini.

3.2. Bounds on mobility

We now provide bounds for signed and symmetric mobility. While net mobility and vertical mobility are panel independent and can be easily measured, symmetric mobility is not and therefore cannot always be directly computed without assumptions on the copula of parent-child ranks. Bounds can nevertheless be informative about the level and trends of symmetric mobility over time. Moreover, bounds also make it possible to compare Lorenz mobility to rank mobility by normalizing their units relative to attainable mobility.

[14] This result parallels the decomposition of income-level mobility in Fields and Ok (1996, Prop. 4.1).

Bounds on signed mobility. It is useful to start from the case of signed rank mobility, as its value is always zero. This happens because ranks are always uniform, regardless of the reshuffling that takes place between generations. The same fact makes horizontal mobility zero-sum for the population as a whole. Take any reassignment (or transport) T of parental into children's ranks, then:

$$\underbrace{\int_0^1 T(r) - r \, dr}_{\text{Net Rank Mobility}} = 0 \qquad \text{and} \qquad \underbrace{\int_0^1 \widetilde{L}(T(r)) - \widetilde{L}(r) \, dr}_{\text{Net Horizontal Mobility}} = 0 \, . \tag{16}$$

This is despite Lorenz mobility pricing the rank changes underlying the offsetting gains and losses differently.

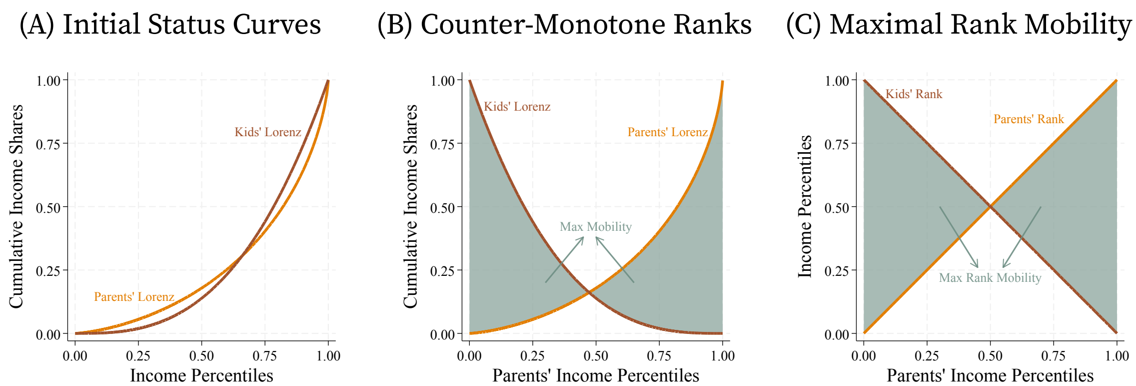
If the Lorenz curve changes across generations, net Lorenz mobility is not necessarily zero because it also contains vertical mobility (Lemma 2), while net rank mobility remains zero. Then, maximal net mobility corresponds to eliminating complete inequality, for a bound of 1/2. If there is information about only one of the generations, the appropriate bound is obtained by eliminating inequality starting from an income distribution with Gini coefficient $\mathcal{G}\left(Y^P\right)$, or going from full inequality to an income distribution with Gini coefficient $\mathcal{G}\left(Y^K\right)$.

Bounds on symmetric mobility. We characterize attainable symmetric mobility in two steps. First, we establish what is the assignment between parent into children's ranks that maximizes mobility. Then, we establish what is the maximal mobility that can be attained under that assignment.

Given two income distributions (Figure 4A), the highest mobility is achieved when dynasties are ordered in counter-monotone fashion, $T(r) = 1 - r$ (Figure 4B),[15] because pair-wise mobility is sub-modular when the Lorenz curve is increasing (when income

[15]The dynasty with the r^{th} highest income in the parent's generation has the $(1 - r)^{\text{th}}$ highest income in the children's generation.

FIGURE 4. Maximal Symmetric Mobility



Notes: Panel A shows Lorenz curves for parents and kids. Panel B shows the maximal average symmetric mobility given these curves obtained by the counter-monotone assignment of parents to kids. Panel C shows the maximal average symmetric rank mobility. Average mobility corresponds to the shaded areas.

is positive). Figures 4B and 4C show this assignment for Lorenz and rank mobility. The counter-monotone assignment therefore provides the following bounds:

$$\overline{m}_R^{\text{sym}} = \int_0^1 |1-2r|\, dr = \frac{1}{2}\,, \qquad \text{and} \qquad \overline{m}_L^{\text{sym}} = \int_0^1 \left|\widetilde{L}^K(1-r) - \widetilde{L}^P(r)\right| dr\,. \tag{17}$$

Changes in inequality that alter the prices of rank reshuffling in Lorenz mobility mean that the second bound can lie above or below 1/2. Similarly, a lower bound is obtained when there is no positional mobility at all and the assignment is $T(r) = r$. In this case, mobility is only vertical:

$$\underline{m}_R^{\text{sym}} = \int_0^1 |r-r|\, dr = 0\,, \qquad \text{and} \qquad \underline{m}_L^{\text{sym}} = \int_0^1 \left|\widetilde{L}^K(r) - \widetilde{L}^P(r)\right| dr\,. \tag{18}$$

These results hold in practical applications as well. When we observe a finite sample, pair-wise mobility is represented by a Monge matrix—a sub-modular discrete function (see Burkard, Dell'Amico, and Martello 2012, Prop. 5.7). Moreover, these bounds do not require parent-child linkages in the data to be computed.

We also obtain a global upper bound by maximizing counter-monotone mobility over the space of Lorenz curves. This is achieved when Lorenz curves are extremal in the sense of Baillo, Carcamo, and Mora-Corral (2022). The upper bound on symmetric Lorenz mobility is[16]

$$0 \leq \int_0^1 \left| L_i^K - L_i \right| di \leq 2 - \sqrt{2}\,. \tag{19}$$

See Appendix B.2 for details.

We illustrate how to use this to compare Lorenz and rank mobility computed from matched administrative data from Norway. Average signed rank mobility is zero by construction, while Lorenz mobility captures intergenerational reductions in inequality. For symmetric mobility, we find Lorenz mobility is over 10 percentage points lower than rank mobility relative to their respective maximum values (Appendix C). This difference reflects how Lorenz ordinates *price* rank mobility by the income shares separating origins and destinations. Inequality compresses income differences across much of the distribution, so most rank changes translate into small changes in Lorenz ordinates.

3.3. Extensions

Aggregating unequal mobility. We further show in Appendix B.1 that aggregate mobility can itself be decomposed into the product of the average mobility and the concentration of that mobility in the population, that is, how unequal mobility is. The result is a representation of aggregate mobility familiar from information-theoretic concentration measures of inequality (Theil 1967) and those that construct equally-distributed-equivalent social welfare measures (Kolm 1969; Atkinson 1970; Sen 1973). In this way, average mobility speaks to the level of mobility in the economy (shaped by the change in inequality) that is then penalized (or augmented) depending

[16] Interestingly, this bound is achievable by horizontal mobility. So, it does not require changes in inequality, instead reflecting how inequality shapes changes in status across positions.

on how concentrated (unequal) mobility is. In effect, mobility concentration provides the equivalent fall in inequality required to exactly offset unequal mobility, completing the link between inequality and mobility. See Atkinson (1970) and Aaberge and Mogstad (2011) for alternative discussions of ordering intersecting Lorenz curves.

Intergenerational Lorenz correlations. An alternative approach to measuring mobility estimates the correlation of individual characteristics across generations. Typically, researchers use intergenerational elasticities of income and the Spearman correlation of income ranks, often obtained from *Galtonian* regressions. The coefficient from log-income regressions can be derived axiomatically as Hart's 1983 mobility measure, as shown in Shorrocks (1993), or as the reduced form of Becker and Tomes (1979) models.

Our approach provides a justification for measuring intergenerational correlations of Lorenz ordinates, instead of income or income ranks. The result is an extension of our analysis to the measurement of the intergenerational correlation or elasticity of status.[17] For example, computing $1 - \operatorname{corr}\left(L_i^P, L_i^K\right)$ or estimating

$$\log\left(L_i^K\right) = \beta_0 + \beta_1 \log\left(L_i^P\right) + u_i\,, \tag{20}$$

where the mobility measure is $1 - \hat{\beta}_1$.

Reassuringly, we find that intergenerational Lorenz correlations are consistent with correlations in ranks and income in practice (Appendix C). Lorenz ordinates and ranks are monotone transformations of income and therefore inherit the same underlying concordance, which produces a common ordering across broad classes of mobility measures (M^c^Gee 2025). Moreover, when inequality is low, the Lorenz curve is approximately linear over much of the distribution, producing intergenerational correlations comparable to those of ranks and log-income.

[17] The term $\log\left(L\left(y_i, Y\right)\right)$ is well-defined as long as $Y > 0$; the same condition required by $\log\left(y_i\right)$. This approach is also motivated as the best linear prediction of status as measured by Lorenz ordinates.

4. Long-run mobility in the United States

We now use our tools to provide new year-by-year measures of inter-generational mobility in the United States for cohorts born between 1915 and 1995. Crucially, measuring net mobility does not require data linking a child to their own parent and can be obtained from the marginal income distributions of parents and children in each cohort. This is a central practical advantage of the framework developed above, and it allows us to measure mobility for cohorts before any reliable intergenerational panel exists.

We find mobility follows a U-shaped pattern and turns negative for cohorts born after 1940. Our results differ from rank-based mobility measures which increase before stabilizing after the 1970s. This reflects how Lorenz mobility prices positional changes across sections of the income distribution.

Our approach, exploiting data from the decennial Censuses and household surveys, differs from the two standard imputation approaches used to extend mobility estimates over long horizons. The first combines linked parent-child data for a subset of cohorts with an imputed copula for the dynasties' joint income distribution (e.g., Chetty, Grusky, Hell, Hendren, Manduca, and Narang 2017; Berman 2022; Manduca, Hell, Adermon, Blanden, Bratberg, Gielen, van Kippersluis, Lee, Machin, Munk, Nybom, Ostrovsky, Rahman, and Sirniö 2024).[18] The second uses proxies for economic status, such as occupational prestige or socioeconomic scores, to approximate parental status and its intergenerational dependence (e.g., Olivetti and Paserman 2015; Ward 2023; Collins and Wanamaker 2022; Jácome, Kuziemko, and Naidu 2025).

[18] Using this approach also makes it possible to compute the concentration of mobility (equation B.9), as well as any signed mobility measure, from cross-sectional information and the implications of the copula for the dependence of ranks between generations.

4.1. Data and sample

This section describes the repeated cross-sections we use, the restrictions that define a parent cohort and a child cohort, and the income concepts we adopt.

Data. We construct marginal income distributions from three repeated cross-sections, all drawn from IPUMS: the decennial Census from 1940 to 2000, the *American Community Survey* (ACS) from 2005 to 2024, and the *Annual Social and Economic Supplement* to the *Current Population Survey* (CPS ASEC) from 1968 to 2024 (Ruggles et al. 2025; Flood et al. 2025). The Census provides large samples reaching back to 1940; the ACS continues that coverage after the discontinuation of the long form following the 2000 Census; and the ASEC supplies annual observations from 1968 that fill the decadal gaps, allowing us to date cohorts to the year rather than the decade. Our sources follow Chetty et al. (2017). We depart from them in extending coverage forward by two decades, in widening the age window for both generations, and—as set out below—in never requiring the copula.

We use self-weighting samples wherever they exist, so that our estimates do not depend on survey weights. In 1940 and 1950 we restrict attention to sample-line persons and their families; from 1990 onward we use the unweighted Census samples. Throughout, we exclude individuals residing in institutional group quarters, which the CPS does not sample, so that the three sources cover a common population. We omit income years 2020 and 2021 from the ASEC and ACS. The suspension of in-person interviewing reduced response rates and changed imputation during the pandemic, introducing significant bias into the distribution of income (Rothbaum and Bee 2021; Parolin, Lehner, and Wilmers 2025).

Sample. A cohort is indexed by the child's year of birth, c. The child cohort consists of individuals born in year c and observed at an age between 30 and 45. The parent cohort consists of individuals who had a child in year c at an age between 16 and 45, and who are themselves observed at an age between 30 and 45. Parents are identified by co-residence with an own child born in year c, so, for each cohort, we pool the survey years in which such a child is young enough to be observed in the household. Where a parent is observed with more than one child born in cohort c, we retain a single child, implying our unit of observation is at the (synthetic) dynasty level.

Three features of this design deserve comment. First, every parent in our sample is observed with a child. Second, we use a common age window rather than a single age. This raises precision in the early cohorts, where the decennial Census is the only source, and it ensures we capture peak life cycle earnings for all cohorts. Third, the decennial structure of the Census before 2005 means that the survey years available for a given cohort, and hence the distribution of parental age at childbearing within the observed parent sample, vary mechanically across cohorts. The resulting pooled dataset has 17,546,722 observations and Appendix D reports the number of observations in each cohort as well as their distribution across data sources.

The repeated cross-sections identify more than population net mobility. They also identify net mobility for any group whose membership is defined consistently across generations; the full distribution of vertical mobility, because the vertical component in equation (3) is a function of the child's rank; and group-average horizontal mobility, which requires only the group-specific marginal distributions in each generation. What the repeated cross-sections do not identify is the dispersion of horizontal mobility within a group, and therefore the concentration of mobility. That is the only object here that requires a parent-child copula, whereas approaches that impute the copula require it throughout (Chetty et al. 2017; Berman 2022; Manduca et al. 2024).

All mobility series below are expressed as percentage points of aggregate income, making their units comparable across cohorts despite substantial income growth. In this, our measures of *relative* mobility complement Ray and Genicot (2023), whose measure of *absolute* mobility captures income growth, by measuring the distributional consequences of growth for individual income positions.

Income measure. Our baseline measure is total family income. Female labour force participation, marriage rates at ages 30 to 45, sorting on education, and the share of business income taxed at the personal level all changed substantially over this period. Parolin, Lehner, and Wilmers (2025) show that these changes have important implications for the measurement of inequality. Focusing on labor earnings for a single focal earner can confound income mobility with changes in family composition and shifts in income sources. Therefore, we compare our baseline results based on family income to results based on individual wage and salary income in Appendix D.[19]

Notably, family income includes business and farm losses and can be negative or zero for the non-employed. While this poses a problem for log-based measures, Lorenz ordinates remain well-defined whenever aggregate income is positive. We retain signed cumulative income shares, allowing the Lorenz curve to fall below zero at the bottom of the distribution.[20]

Groups. We use three race groups to disaggregate population mobility: White, Black, and non-White non-Black.[21] Cells for this third group are very small before 1955, and the Hart–Celler Act of 1965 changed its composition. Consequently, the series would mix mobility with changes in group membership. Appendix Figure D.5 reports results for this third group. We classify adult children by place of birth rather than residence

[19] A related concern is that survey coverage of income concepts changes over time. The individual wage and salary series in Appendix D provides a narrower income concept that can be compared with the baseline. The focal-earner series has the same qualitative pattern as the baseline, although its mid-century decline is larger (Figure D.2C).

[20] Ultimately, negative incomes are quantitatively unimportant in our analysis. A related concern is income topcoding: we use the Census Bureau's rank-proximity-swap cell means for the ASEC from 1976 onward and internal topcodes elsewhere. As we discuss above, the thinness of the right tail of the income distribution mitigates the impact of this measurement error.

[21] Our choice of three groups reflects the need to harmonize different data sources over a long horizon.

when grouping individuals by region, so that the regional classification is not itself an outcome of migration. We compare children's birthplace with parents' adult residence. Because harmonized birthplace is available only in the Census, the geographic results use Census observations.

Inference. For expositional simplicity, we report point estimates in the main text. We construct confidence intervals using a stratified bootstrap procedure, stratifying by year and underlying data source and resampling at the household level within strata. Confidence intervals are generally narrow. Appendix D.1 reproduces our main results along with their 95% confidence intervals.

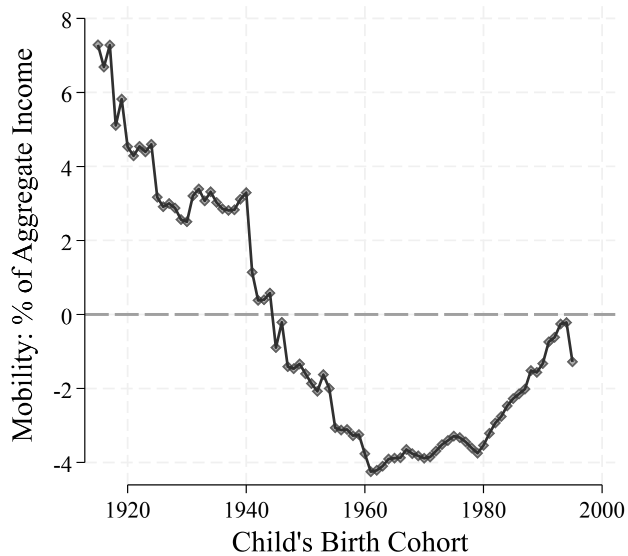
4.2. Long-run net mobility in the U.S.

We measure net Lorenz mobility for 81 annual birth cohorts born between 1915 and 1995, and decompose the series by race and region. Because net mobility is panel independent, the scarcity of linked intergenerational panels is not a binding constraint.

Figure 5 reports net mobility by birth cohort. Across the first half of the sample it falls by roughly 11 percentage points of aggregate income before recovering in a U-shaped pattern. The cohort born in 1915 experiences positive net mobility equivalent to being closer to the top of the income distribution than their parents by 7 percent of aggregate income. Mobility then declines steadily, crossing zero for cohorts born in the early 1940s, and reaches a trough of approximately −4 percent for cohorts born around 1960. It remains near that level through the 1960–1980 cohorts before recovering toward zero for cohorts born in the late 1980s and early 1990s.

Thus, the long-run series contains three distinct periods: a four-decade decline for cohorts born before 1960, a two-decade plateau of negative mobility, and a partial recovery. Crucially, the plateau is not a plateau at neutrality. The average child born between 1960 and 1980 occupies a position worth roughly 4 percentage points of

FIGURE 5. Long-run Net Mobility



Notes: The figure reports average signed mobility in Lorenz ordinates, as defined in equation (11), by the child's birth cohort. The series is the difference between the average Lorenz ordinate of the child and parent marginal family-income distributions and is expressed in percentage points of aggregate income.

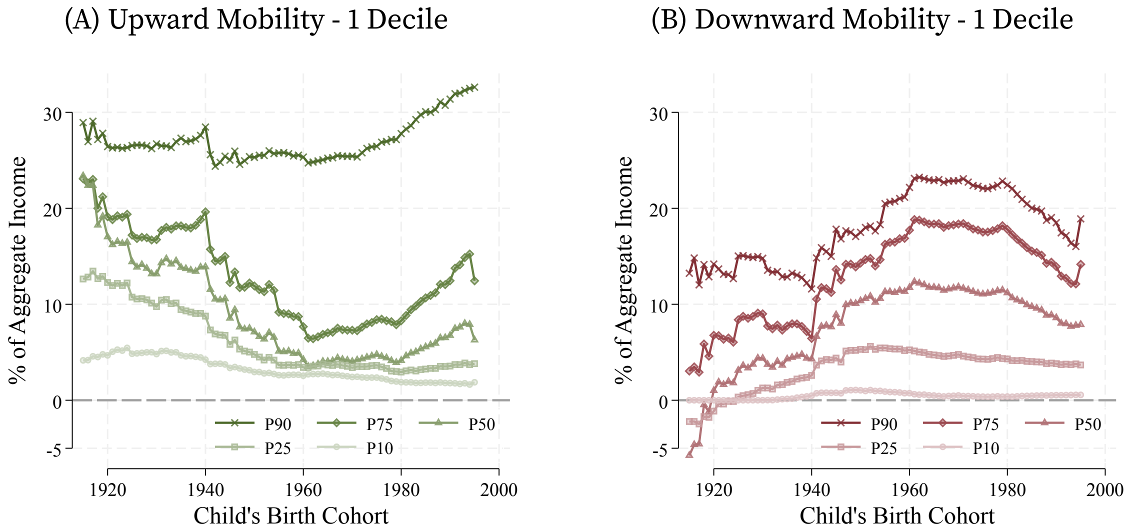
aggregate income less than the position occupied by their parents.[22] These results on relative mobility contrast with rising average incomes over the 20th century but align with the decline in lifetime incomes and increase in lifetime income inequality among American men born 1942 to 1958 (Guvenen et al. 2022). Appendix Figure D.2 shows our results are robust to changes in the income measure, age brackets, and data source. Figure D.8A shows changes in mobility are statistically significant.

Pricing mobility. Why does this decline differ from historical rank-based evidence? Relative (rank) mobility generally rises for the cohorts born between the 1910s and 1940s (Jácome, Kuziemko, and Naidu 2025; McGee 2025).[23] For later cohorts, the trends in

[22]For reference, Norwegians born between 1960 and 1970 were, on average, 1.6 percentage points of income closer to the top of the distribution than their parents. This grows to 3.1 percentage points once taxes and transfers are taken into account. See Appendix C.

[23]The rank-rank slope decreases over the same period in which net Lorenz mobility falls to its trough. However, not all rank-based studies find this pattern. Davis and Mazumder (2024) document a sharp decline in relative mobility between the cohorts born around 1950 and those born around 1960.

FIGURE 6. Pricing Rank Mobility



Notes: Panel A reports the change in Lorenz ordinates associated with moving up one decile, from starting percentiles 10, 25, 50, 75, and 90. Panel B reports the corresponding magnitude of a one-decile downward move. Values are percentage points of aggregate income; a negative value in Panel B means that the upward vertical shift in the Lorenz curve more than offsets the downward rank movement.

the two measures agree: percentile rank-based measures are nearly constant for the 1971–1993 birth cohorts (Chetty et al. 2014), just as net mobility is flat for the cohorts born between 1960 and 1980. Nevertheless, our estimates suggest mobility stabilized at a negative value, reflecting net losses relative to parent generations.

The divergence in the earlier period reflects cardinal information captured by Lorenz mobility. The change in Lorenz ordinates for each dynasty combines positional changes with changes in inequality, *pricing* rank movements according to the income shares separating their origins and destinations. As inequality changes, the pricing of rank movements also changes. Consequently, positional changes lead to net mobility.[24] Ranks hold that price fixed at one by construction, holding ordinal but not cardinal information and implying zero net mobility.

Figure 6 makes the changes in price explicit for one-decile moves. As in Lemma 1,

[24]In contrast, horizontal mobility, which keeps the pricing schedule of positional moves constant, is zero-sum as discussed in Section 3.2.

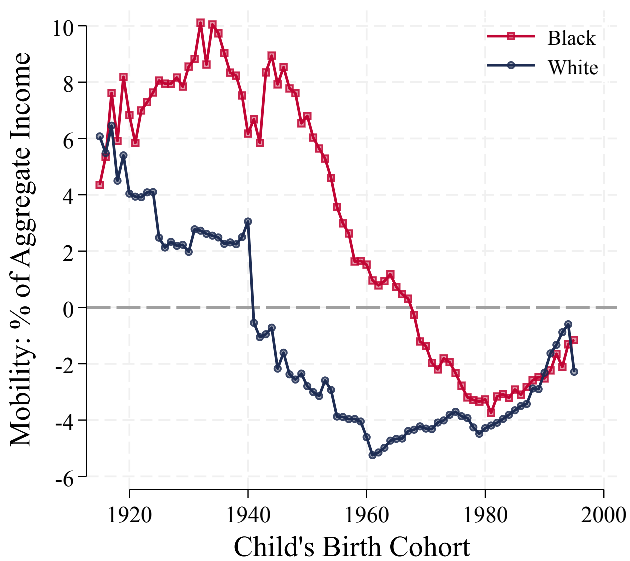
an upward move near the top of the distribution is worth more aggregate income than the same move near the bottom (Panel 6A). For the earliest cohorts, a move from the 50^{th} to the 60^{th} percentile is worth roughly 15 percentage points of aggregate income, compared with 25–30 percentage points for a move from the 90^{th} percentile to the top. However, the value of upward movements decreases over time, except at the top. By 1960, a movement from the median is worth 4 percentage points, while the move at the top is essentially unchanged. This limits the effect on relative mobility of upward positional moves in the bottom half of the distribution.

Panel B shows the mirror image for downward moves, pricing one-decile falls. The magnitude of the loss grows through the middle of the century for almost all starting positions, most sharply near the top. An important difference is for downward mobility at the bottom, as a further decrease from the 10^{th} percentile does not represent a large change in terms of income throughout the entire sample. Moreover, the role of the initial decrease in overall inequality is seen for the earliest cohorts: a one-decile fall for dynasties below the median can nevertheless register as a gain up to the early 1920s, because the compression of the income distribution raises the child's Lorenz curve by more than the positional loss subtracts.

This has practical content. Over time, going from the 1915 to the 1960 cohort, most upward moves are priced less, and downward moves priced more. These changes in prices lead to net declines in the average position of each cohort relative to their parents after 1940. Net mobility starts to recover after 1980 when upward moves start being priced more highly and downward moves entail a lower loss in income shares.

Mobility of Black and White Americans. The aggregate series masks sharply different racial trajectories that we document in Figure 7. Net mobility for White Americans follows the population pattern: positive for the earliest cohorts, falling steadily, and negative after 1940. For Black Americans it instead rises for cohorts born through the

FIGURE 7. Long-run Net Mobility by Race



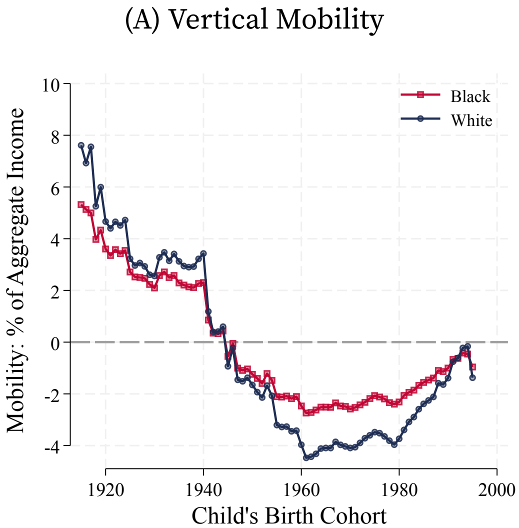
Notes: The figure reports average signed mobility in Lorenz ordinates separately for Black and White Americans, by the child's birth cohort. Group mobility is the average change in aggregate-income shares for members of that group, as characterized in Corollary 2.

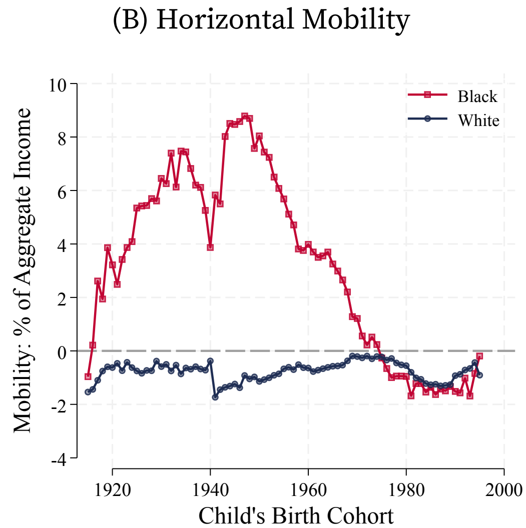
1930s, holds at roughly 8 percentage points of aggregate income through cohorts born in the 1950s, and declines thereafter. The two series converge only for cohorts born in the 1980s and 1990s. This path is consistent with historical evidence of substantial Black–White positional convergence during the twentieth century (Ward 2023; Collins and Wanamaker 2022; Jácome, Kuziemko, and Naidu 2025).[25]

The decomposition in Figure 8 explains why the mobility of Black and White Americans differs for most of the sample period. The vertical component—the change attributable to the shape of the income distribution, holding each group's position in it fixed—declines for both groups and, while comparable throughout, is more negative for White Americans after 1960, consistent with White Americans having higher average positions in the income distribution. The horizontal component—the change attributable to the group's movement across the population distribution—is small and

[25] Convergence in unconditional net mobility does not imply convergence conditional on parent income.

FIGURE 8. Decomposing Long-run Net Mobility by Race





Notes: The figure decomposes the group-specific net mobility in Figure 7 into the vertical and horizontal components defined in equation (3). Panel A holds each group's position in the population distribution fixed and reports the change associated with the parent and child Lorenz curves. Panel B reports the change associated with the group's movement across the population distribution.

negative for White Americans throughout the sample, but strongly positive for Black Americans, rising for cohorts born through the 1940s, before falling. The early Black–White gap in net mobility is therefore almost entirely horizontal, resulting from Black Americans' rise through the distribution, and its subsequent closing is the unwinding of that horizontal advantage rather than a change in the price of the same positional mobility.[26]

This distinction between vertical and horizontal mobility also reconciles our estimates with the historical rank-based and income-based evidence. The rise of horizontal mobility for Black Americans through the middle of the sample is consistent

[26]The same two forces can also be distinguished within generations. Bayer and Charles (2018) distinguish movement across the male earnings distribution from changes in the distribution itself. Their positional force is the within-generation counterpart of our horizontal component, and their distributional force is the counterpart of our vertical component. Two differences are worth stating. First, Bayer and Charles compare two separately constructed objects—a level gap and a rank gap—whereas the vertical and horizontal terms here are components of a single measure and sum to net mobility by construction. Second, their comparison is between contemporaneous groups while ours is between children and their parents.

with large rank gains through the cohorts born in the 1970s (as we show in Appendix Figure D.3). These gains from upward positional mobility wind down as we see substantial positional convergence between Black and White Americans. This pattern of large rank gains for Black Americans followed by convergence did not materialize in convergence in incomes. Our results show convergence in mobility took place against a backdrop of negative aggregate net mobility, in which the average member of both groups loses ground relative to their parents: Black Americans moved up the distribution over a period in which the distribution was stretching at the top and compressing at the bottom, so that upward rank movements meant smaller (relative) income gains except at the top (Figure 6A).[27]

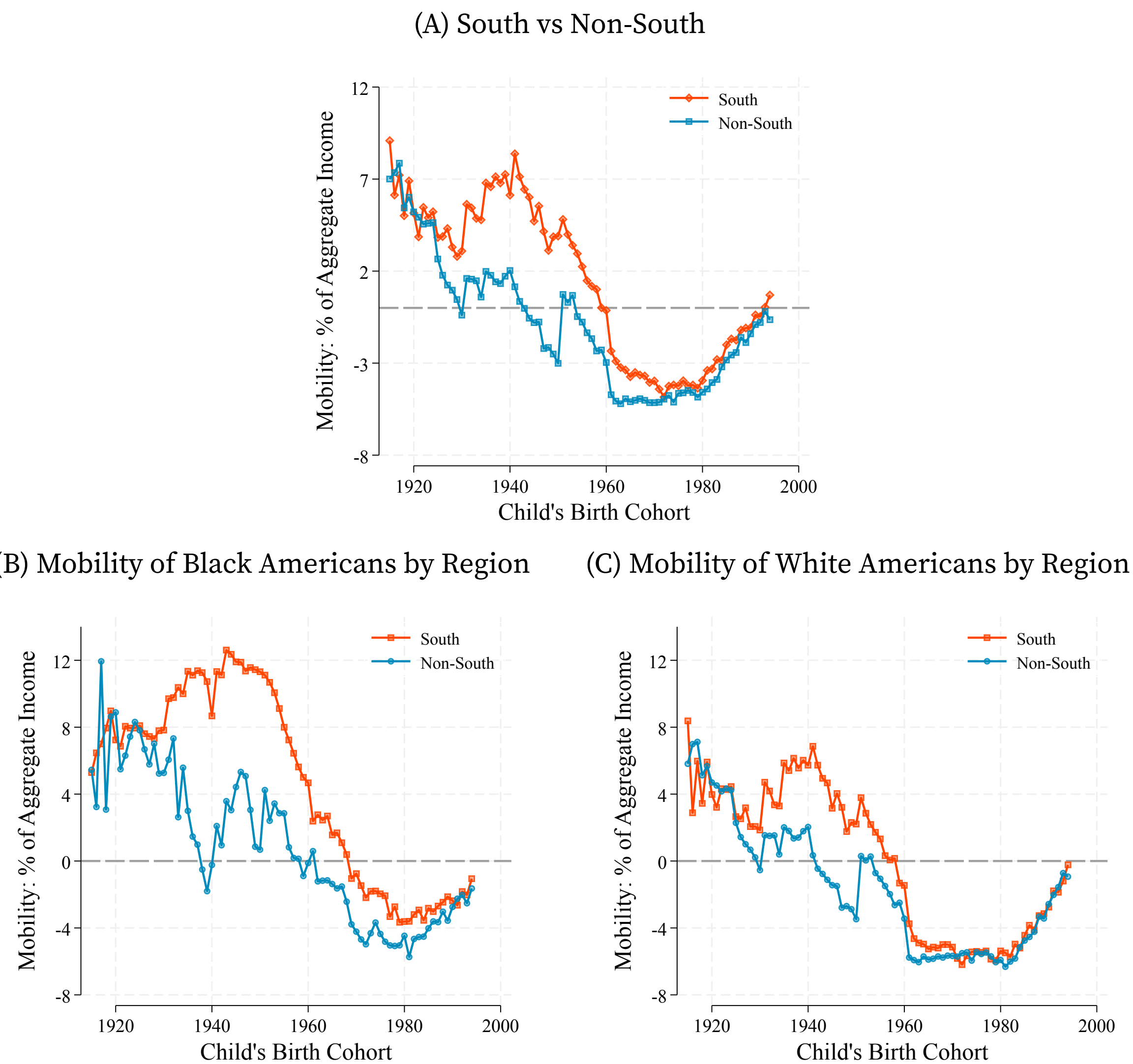
The negative mobility we measure for Black Americans after 1970 is also consistent with Chetty et al. (2020), who document that Black Americans in these later cohorts have both lower rates of upward mobility and higher rates of downward mobility than White Americans conditional on parental position. Despite these differences, White Americans experience lower net mobility because losses near the top of the distribution become more expensive over the century (Figure 6B), while Black Americans on average experience moves in more compressed regions of the income distribution.

The geography of net mobility. Racial differences map onto regional ones. Figure 9 separates children born in the Census South region from those born elsewhere.[28] Children born in the South experience greater net mobility for the early cohorts, but

[27]Manduca (2018) documents the same tension in family income: the Black–White gap in family income rank narrowed by almost one third between 1968 and 2016, while the ratio of median Black to median White family income was constant. The corresponding within-group mobility series in Appendix Figure D.3A is also much lower for Black Americans, confirming that most of the gain in the population distribution reflects convergence between groups rather than a broad improvement within the income distribution of Black Americans.

[28]Appendix Figure D.7 further separates the Non-South Census regions. There are differences in the timing and magnitude of the mobility trends. The Midwest starts with a lower level of mobility and has a more pronounced decline. The Northeast is the first to exhibit negative mobility and remains relatively stable at negative values after 1925. Finally, the West follows the overall mobility trend already described.

FIGURE 9. The Geography of Mobility



Notes: The figure reports average signed mobility in Lorenz ordinates by the child's region of birth. Panel A compares children born in the Census South region with children born elsewhere. Panels B and C make the same comparison separately for Black and White Americans. Parents are classified by adult residence, while children are classified by birthplace.

the regional gap closes around mid-century. Panels B and C show this difference is concentrated among Black Americans. Mobility for Black Americans born in the South is much higher than for those born elsewhere through the cohorts born in the 1950s, whereas the regional series for White Americans are close to one another after 1960.

Migration is one possible contributor to these descriptive regional differences.

Appendix Figure D.6 separates adult children who remain in their birth region from those who leave it. Leavers experience greater mobility than stayers, especially among Black Americans born in the South. Regional differences in Black mobility have general-equilibrium origins as well as compositional ones. Derenoncourt (2022) shows racial composition shocks in northern destinations during the Great Migration eroded the returns to growing up outside the South for Black families, accounting for 27 percent of the region's racial upward mobility gap today. Our estimates cannot separate these channels, but the timing of net mobility coincides with them. Because migration is an outcome and the parent and child observations are not linked, these estimates should not be interpreted as the causal effect of moving.

5. Conclusions

Our approach to measuring mobility, by adopting Lorenz ordinates, has several advantages for applied research. It directly addresses the concern that purely ordinal measures may overstate mobility when positional changes correspond to small income differences. Lorenz mobility brings standard concepts from the study of inequality to the study of mobility and incorporates both ordinal and cardinal information. In unequal societies, differences in Lorenz ordinates are small where the income distribution is compressed and large where income is dispersed, capturing material differences between individuals.

Additionally, the net change in Lorenz ordinates can be measured without panel data. This property is shared by vertical mobility and signed horizontal mobility. This makes it possible to measure the average change in position relative to the top (or bottom) of the income distribution. Moreover, the Lorenz curve has a long history of statistical interest and many techniques exist to reconstruct the Lorenz curve from coarse groups and can be extended to mobility in education or other proxies and outcomes. It also downweights the impact of measurement error across much of the distribution. As with ranks, measurement error is more important where the density is highest, but the properties of the Lorenz curve make measurement error less important outside of the right tail—typically, a thin region of the income distribution. These advantages make a Lorenz-based approach particularly attractive in settings where data collection is challenging.

Our long-run U.S. application illustrates the value of panel independence. Using repeated cross-sections, we show net mobility declined from positive values for the cohorts born in the 1910s, became negative after 1940, and partially recovered for cohorts born after 1980. The divergence from historical rank-based estimates reflects changes in the income attached to positional movements. The same distinction is central to understanding differences in mobility between Black and White Americans: Black Americans made substantial gains in their position in the income distribution, but those gains translated into more limited income changes.

Our results show mobility is negative for Americans born after the 1940s. Each generation loses ground relative to their parents, increasing their income distance to the top and reducing their distance to the bottom. This happens even as positional mobility increases, a potential explanation for why the public perception of mobility is pessimistic (Alesina, Stantcheva, and Teso 2018) and differs from measures of overall mobility and economic activity.

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Appendix A. Ranks and Lorenz Ordinates as Economic Status

We consider an economy populated by a finite set of dynasties indexed $i = 1, \ldots, N > 2$. There are two observations of income for each dynasty, y^P and y^K; these might be incomes of parents and their children or of the same individual at different dates. We denote by Y^P and Y^K the $(N \times 1)$ vectors of income. We define functions r and L as the rank and the Lorenz curve of Y, respectively,

$$r(y, Y) \equiv \sum_{i=1}^{N} \frac{\mathbb{1}_{\{y_i \leq y\}}}{N} \quad \text{and} \quad L(y, Y) \equiv \sum_{\{\tilde{y} \in Y \mid \tilde{y} \leq y\}} \frac{\tilde{y}}{N\mu}; \quad \text{where} \quad \mu \equiv \sum_{i=1}^{N} \frac{y_i}{N}. \tag{A.1}$$

For simplicity of exposition, we assume no ties in the distribution of income. This is the case in the continuous setting of the main text when the distribution of income admits a continuous density.

We give a justification for the use of Lorenz ordinates in the study of mobility by providing conditions under which Lorenz ordinates capture the *status* of individuals in society. We view status as capturing the consumption capabilities of individuals (e.g., Sen 1973; Yitzhaki 1979) as well as their position in the economy. Thus, we assume status is a function of an individual's income and the income distribution in the economy, $s : \mathbb{R}_+ \times \mathbb{R}_+^N \longrightarrow \mathbb{R}_+$.

We constrain how status depends on income by imposing 4 conditions expressed in Axioms A1–A4 below. We start by normalizing the scale of status, forcing it to be bounded.

Axiom A1 (**Status Normalization**). *Status satisfies* $s(y, Y) \in [0, 1]$.

We restrict attention to status functions that are strictly monotone in income. Monotonicity is key to distinguishing purely ordinal representations of status, like income ranks, from cardinal representations that reflect income levels. Specifically, status increases with the dynasty's income, holding the rest of the income distribution constant.

Axiom A2 (**Income Monotonicity**). *Status* $s\left(y_i, Y\right)$ *is strictly increasing in* y_i *at interior ranks,* $r_i < 1$.

In order to disentangle economic growth from mobility, we focus on status functions that capture *relative*, as opposed to *absolute*, mobility. Therefore, status depends on the shape of Y and the relative position of y in Y, but not on their level.

Axiom A3 (**Growth Independence**). *Status satisfies* $s(y, Y) = s(\kappa y, \kappa Y)$ *for any* $\kappa > 0$.

Finally, we establish how status responds to the shape of the income distribution. Our objective is that status combines the cardinal and ordinal information reflected in income differences. Specifically, we make status *aspirational*: dynasties aspire to be ahead of others and reach those above. Status increases when the dynasty gets closer (in income) to those above and further away from those at the bottom of the distribution, given their position in the distribution. We strengthen this by imposing that status be invariant to pure redistribution above or below the dynasty. This makes status "*anonymous*:" a dynasty does not care about the identities of others (whose positions can be reshuffled by redistribution) while still caring about their own relative position (unchanged by this redistribution). This property makes the cumulative distribution of income the relevant statistic for determining status.

AXIOM A4 (**Aspirational Status**). *Order dynasties in Y such that $y_i \leq y_{i+1}$ for all $i = 1, \ldots, N$. Let $1 < h < j \leq N$ be two dynasties and make any transfer δ between them such that*

$$y_{h-1} < y_h + \delta < y_{j+1} \qquad \text{and} \qquad y_{h-1} < y_j - \delta < y_{j+1} \tag{A.2}$$

letting $y_{j+1} = y_j$ if $j = N$. The transfer δ can be positive or negative. Then,

(a) *The status of dynasties $i = 1, \ldots, h-1, j+1, \ldots, N$ remains unchanged: $s(y_i, Y) = s\left(y_i, Y^{\delta}\right)$ where Y^{δ} is the same as Y except for the incomes of the h and j entries; and*

(b) *The status of dynasty j is unchanged if their rank is unchanged: if $y_{j-1} < y_j - \delta < y_{j+1}$, then $s\left(y_j, Y\right) = s\left(y_j - \delta, Y^{\delta}\right)$.*

Together, these axioms tie status, and hence mobility, to the Lorenz ordinates of dynasties in the income distribution as well as to their income ranks. Status reflects cardinal differences in income, because increasing income brings a dynasty closer to those they aspire to reach, even when income ranks remain unchanged.

PROPOSITION A1. *A status function satisfies Axioms A1–A4 if and only if*

$$s(y, Y) = \psi\left(r(y, Y), L(y, Y)\right), \tag{A.3}$$

where $\psi(r, L) \in [0, 1]$ for all feasible rank–Lorenz-ordinate pairs and ψ is strictly increasing in its second argument for every feasible rank r holding Y_{-i} fixed (with this condition vacuous at $r = 1$), replacing y_i with $y_i' > y_i$ so that $r\left(y', Y'\right) > r(y, Y)$, implies $\psi\left(r\left(y', Y'\right), L\left(y', Y'\right)\right) > \psi(r(y, Y), L(y, Y))$, so that status increases.

PROOF. We first prove necessity. Let Y be such that $y_i \leq y_{i+1}$ for all $i = 1, \ldots, N$ and fix a dynasty i. Consider an alternative $N \times 1$ ordered income vector Y' with the same average income, $\mu' = \mu$, and cumulative income among those richer than i, $\sum_{\{\tilde{y}' > y_i'\}} \tilde{y}' = \sum_{\{\tilde{y} > y_i\}} \tilde{y}$. Assume the vectors coincide in their i^{th} entry, $y_i' = y_i$, so the sum of income below i also coincides.

It is possible to move from Y to $Y^{'}$ by means of transfers among dynasties $1, \ldots, i-1$ and dynasties $i+1, \ldots, N$. Axiom A4 demands $s(y_i, Y) = s\left(y_i, Y^{'}\right)$. Therefore, status cannot depend on the whole distribution of income but only on y_i, its rank, r, mean income, μ, and the sum of income above y_i, $\sum_{\{\tilde{y}>y_i\}} \tilde{y}$.

A transfer from i to $h < i$ that does not change i's rank does not change status (Axiom A4). So, status cannot depend on y_i directly. Equivalently, it can depend on y_i and the incomes below it only through the cumulative income sum $\sum_{\{\tilde{y}\leq y_i\}} \tilde{y}$, which is determined by mean income and the income above y_i.

Axiom A3 implies μ is superfluous—re-scaling income results in the same status—so we are left with two arguments (ranks and cumulative income shares) and write $s(y, Y) = \psi\left(r(y, Y), \sum_{\{\tilde{y}\leq y\}} \frac{\tilde{y}}{N\mu}\right)$ for some function ψ. The second argument is the Lorenz ordinate.

We now show that ψ must be strictly increasing in the Lorenz ordinate. This follows directly from the fact that an increase in income that holds ranks constant increases the Lorenz ordinate of the dynasty by definition (except when $r = 1$). Then, ψ must be strictly increasing in L for a given r to satisfy Axiom A2. The second condition follows from the same Axiom when income changes imply rank changes.

We finally prove sufficiency. Let s be given by (A.3), with ψ satisfying the two stated properties. Its range gives Status Normalization, while the scale invariance of both ranks and Lorenz ordinates gives Growth Independence. The transfers in Axiom A4 preserve total income and preserve the relevant dynasty's rank and cumulative income through that rank; hence, they preserve its Lorenz ordinate and its status. Thus, Aspirational Status is satisfied.

To verify Income Monotonicity, hold all other incomes fixed and increase dynasty i's income from y to $y' > y$, starting from an interior rank. If its rank is unchanged, the Lorenz ordinate strictly increases, so the first property of ψ implies that its status strictly increases. If its rank increases, the same conclusion follows from the technical feasibility property when ranks change. Therefore, all four axioms are satisfied. □

We further isolate how status depends on the shape of the income distribution by restricting how it moves across co-monotone distributions. Specifically, we require status moves linearly along the ray connecting any two such distributions. This condition sets how status units deal with the cardinality of income, as the position (rank) of each dynasty is fixed along these movements.

AXIOM A5 (**Co-Monotone Rays**). *Order dynasties in Y such that $y_i \leq y_{i+1}$ for all $i = 1, \ldots, N$. Consider an alternative income vector $Y^{'}$ with the same strict order for all dynasties. An* *economy along the ray connecting these two vectors has* $Y^\alpha = \alpha Y + (1-\alpha) Y'$, *for* $\alpha \in [0,1]$. *Status satisfies* $s\left(y_i^\alpha, Y^\alpha\right) = \alpha s\left(y_i, Y\right) + (1-\alpha) s\left(y_i', Y'\right)$.

Axiom A5 only covers mixing of co-monotone income distributions. The outcome of transfers that change the rank of a dynasty is not covered; neither are distributions with rank reversals between dynasties. Among co-monotone distributions, we impose no restrictions on the shape of Lorenz curves or the change in status levels between Y and Y'. In particular, Lorenz curves can cross, allowing $s\left(y_i, Y\right) > s\left(y_i', Y'\right)$ for some i and $s\left(y_j, Y\right) < s\left(y_j', Y'\right)$ for some $j \neq i$. Nevertheless, Axiom A5 is enough to obtain that the ψ function in (A.3) is affine in Lorenz ordinates.

PROPOSITION A2. *A status function satisfies Axioms A1–A5 if and only if*

$$s\left(y_i, Y\right) = \Omega\left(r_i\right) \cdot \left[\lambda L_i + (1-\lambda) \Xi\left(r_i\right)\right], \tag{A.4}$$

for a positive function $\Omega(r) > 0$, *a nonnegative function* $\Xi(r) \geq 0$, *and a scalar* $\lambda \in (0,1]$, *where* $r_i = r\left(y_i, Y\right)$ *and* $L_i = L\left(y_i, Y\right)$ *are the rank and Lorenz ordinate of dynasty* i. *The right-hand side of* (A.4) *belongs to* $[0,1]$ *for every feasible rank–Lorenz-ordinate pair and satisfies the feasible rank-change condition in Proposition A1.*

PROOF. Let Y be such that $y_i \leq y_{i+1}$ with strict inequality for at least one i, and let Y' be an income vector with the same strict order. Without loss, we have $\mu = \mu'$, as status and the Lorenz curves are scale invariant (Axiom A3).[1] Under Proposition A1, Axiom A5 requires

$$\begin{aligned} s\left(y_i^\alpha, Y^\alpha\right) &= \psi\left(r\left(y_i^\alpha, Y^\alpha\right), L\left(y_i^\alpha, Y^\alpha\right)\right) \\ &= \alpha\psi\left(r\left(y_i, Y\right), L\left(y_i, Y\right)\right) + (1-\alpha)\psi\left(r\left(y_i', Y'\right), L\left(y_i', Y'\right)\right). \end{aligned} \tag{A.5}$$

From Aaberge (2001), the Lorenz curve is linear under co-monotone mixtures,

$$L\left(y_i^\alpha, Y^\alpha\right) = \alpha L\left(y_i, Y\right) + (1-\alpha) L\left(y_i', Y'\right). \tag{A.6}$$

[1]Formally, define $\widehat{Y}' \equiv (\mu/\mu')Y'$ and $\widehat{y}_i' \equiv (\mu/\mu')y_i'$ for every dynasty i. The rescaled vector $\widehat{Y}'$ has mean μ and the same ordering and Lorenz curve as Y', so $r(\widehat{y}_i', \widehat{Y}') = r(y_i', Y')$ and $L(\widehat{y}_i', \widehat{Y}') = L(y_i', Y')$. Growth Independence also implies $s(\widehat{y}_i', \widehat{Y}') = s(y_i', Y')$. We therefore apply Axiom A5 to the ray connecting Y and $\widehat{Y}'$ and, with a slight abuse of notation, continue to denote the rescaled vector by Y'. Equivalently, when the original vectors have different means, define $\widehat{\alpha} \equiv \alpha\mu/[\alpha\mu+(1-\alpha)\mu']$. The original mixture $\alpha Y+(1-\alpha)Y'$ is proportional to the mean-normalized mixture $\widehat{\alpha}(Y/\mu) + (1-\widehat{\alpha})(Y'/\mu')$, so Growth Independence allows the ray to be parameterized by $\widehat{\alpha}$.

Hence, ψ satisfies (A.5) if and only if it is affine in L at every feasible rank, so write

$$\psi(r, L) = a(r)L + b(r). \tag{A.7}$$

Proposition A1 implies $a(r) > 0$ at every rank $r < 1$. Status Normalization implies $b(r) \geq 0$, since at every such rank feasible Lorenz ordinates can be arbitrarily close to zero. At rank one, $L = 1$ is the only feasible Lorenz ordinate, so the two coefficients are not separately identified. Income Monotonicity and Status Normalization imply $\psi(1, 1) > 0$, so they can be chosen with $a(1) > 0$ and $b(1) \geq 0$ to reproduce $\psi(1, 1)$.

Fix any scalar $\lambda \in (0, 1)$ and define

$$\Omega(r) \equiv \frac{a(r)}{\lambda} > 0 \qquad \text{and} \qquad \Xi(r) \equiv \frac{b(r)\lambda}{a(r)(1-\lambda)} \geq 0. \tag{A.8}$$

Substitution into (A.7) gives (A.4). The special case $\lambda = 1$ is also available when $b(r) = 0$ at every rank.

Conversely, suppose status has the form (A.4) and satisfies the stated range and feasible rank-change conditions. Its coefficient on L is $\lambda\Omega(r) > 0$, so Proposition A1 implies that Axioms A1–A4 are satisfied. Along a co-monotone ray, ranks remain fixed and Lorenz ordinates satisfy (A.6); because (A.4) is affine in L at each rank, it then satisfies Axiom A5. □

Thus, status depends on the dynasties' Lorenz ordinates—their positions in the Lorenz curve—and not just in their ranks—their positions in the income distribution. This endows status with cardinal information, sets the units and range of status, and makes explicit the link between inequality and status (as discussed in Section 2). The nonnegative weights on the Lorenz ordinates can capture social preferences that depend on the position of individuals, reinforcing the role of inequality.[2] Finally, while our representation of status goes beyond ranks, they are a limiting case when $\lambda \longrightarrow 0$, $\Omega(r) = 1$, and $\Xi(r) = r$. When $\lambda = 1$ and $\Omega(r) = 1$, status coincides with Lorenz ordinates.

Appendix B. Mobility Measures

We provide a standard characterization of status mobility taking the measurement of economic status as given. First, we impose that mobility be symmetric, thus it measures the absolute change in status, similar to Fields and Ok (1996) for income levels.

Proposition A3 (**Absolute Status Mobility**). *A mobility function* $m : \mathbb{R}_+ \times \mathbb{R}_+ \longrightarrow \mathbb{R}_+$ *is*

[2] This is consistent with rank-based weights in Yaari's 1988 evaluation measure, the parametric weighting in Donaldson and Weymark's 1980 S-Gini index, as well as generalized concentration indexes (Kakwani, Wagstaff, and Van Doorslaer 1997, Wagstaff 2002).

(a) *Continuous in all arguments;*

(b) *Symmetric in that* $m\left(s, s'\right) = m\left(s', s\right)$*;*

(c) *Step-additive in that* $m\left(s, s''\right) = m\left(s, s'\right) + m\left(s', s''\right)$ *for all* $s \leq s' \leq s''$*; and*

(d) *Translation-invariant in that* $m\left(s+\zeta, s'+\zeta\right) = m\left(s, s'\right)$ *for all* s, s' *and* $\zeta > 0$*;*

if and only if

$$m\left(s^P, s^K\right) \propto \left|s^K - s^P\right| . \tag{B.1}$$

Proof. Consider status s. By translation invariance $m(s, s+\zeta) = m(0,\zeta)$ for $\zeta \geq 0$. So, mobility does not depend on s, only on the gap ζ. Define mobility by a gap ζ as $g(\zeta) \equiv m(0,\zeta)$. g is immediately continuous because m is continuous. Moreover, mobility is additive on the gap, that is, $g\left(\zeta+\zeta'\right) = g(\zeta) + g\left(\zeta'\right)$. To see this, let $\zeta \geq \zeta' \geq 0$. Step-additivity of m gives

$$g\left(\zeta+\zeta'\right) = m\left(0, \zeta+\zeta'\right) = m(0,\zeta) + m\left(\zeta, \zeta+\zeta'\right) = g(\zeta) + g\left(\zeta'\right) . \tag{B.2}$$

Therefore, $g(\zeta) = \kappa\,\zeta$ for some $\kappa \geq 0$, because continuous additive functions are linear.

Consider s and s' with $s \leq s'$. We have $m\left(s, s'\right) = g\left(s'-s\right) = \kappa(s'-s)$. Symmetry extends the characterization to all status pairs s and s' with $m\left(s, s'\right) = \kappa\,|s'-s|$. □

Second, we replace symmetry in favor of *signed-symmetry*.

Proposition A4 (**Signed Status Mobility**). *A mobility function* $m : \mathbb{R}\times\mathbb{R} \longrightarrow \mathbb{R}$ *satisfies properties (a), (c), and (d) of Proposition A3 and is signed-symmetric, in that* $m\left(s, s'\right) = -m\left(s', s\right)$ *with* $m\left(s, s'\right) > 0$ *for all* $s < s'$*, if and only if*

$$m\left(s^P, s^K\right) \propto s^K - s^P . \tag{B.3}$$

Proof. Consider status s and s' with $s < s'$. We have $m\left(s, s'\right) = g\left(s'-s\right) = \kappa(s'-s) > 0$ from the proof of Proposition A3. Signed-symmetry extends the characterization to all status pairs s and s' with $m\left(s, s'\right) = \kappa\left(s'-s\right)$ and $\kappa > 0$. □

We now turn to aggregating mobility into a single statistic. Formally, we aggregate mobility via a function $\mathcal{M} : \mathbb{R}^N \longrightarrow \mathbb{R}$ from a vector M of dynastic mobilities, with a typical element m_i. We adopt the Kolmogorov-Nagumo-de Finetti's theorem axiomatic characterization of *quasi-arithmetic* aggregators (Hardy, Littlewood, and Pólya 1952, Thm. 215).

THEOREM A1 (**Kolmogorov-Nagumo-de Finetti**). *A family of aggregators $\mathcal{M}_N : \mathbb{R}^N \longrightarrow \mathbb{R}$, indexed by N, is*

(a) *Continuous and strictly increasing in all arguments;*

(b) *Symmetric in that $\mathcal{M}_N(PM) = \mathcal{M}_N(M)$ for any permutation matrix P;*

(c) *Reflexive in that $\mathcal{M}_N(m\mathbb{1}_N) = m$ for any m; and*

(d) *Associative in that $\mathcal{M}_N(M) = \mathcal{M}_N([\bar{m}, \ldots, \bar{m}, m_{K+1}, \ldots, m_N])$, where $\bar{m} = \mathcal{M}_K([m_1, \ldots, m_K])$;*

if and only if there is a continuous function Γ such that

$$\mathcal{M}_N(M) = \Gamma^{-1}\left(\frac{1}{N}\sum_{i=1}^{N}\Gamma(m_i)\right). \tag{B.4}$$

These aggregators encompass standard additive aggregators common in the mobility literature (e.g., Bartholomew 1973; Fields and Ok 1996, 1999; Cowell and Flachaire 2018; Ray and Genicot 2023). They respect monotonicity, so that higher mobility for any dynasty is reflected in higher aggregate mobility, and anonymity, so the indexes of dynasties are superfluous. They also satisfy an *associative* property, so that $\mathcal{M}_N(M) = \mathcal{M}_N([\overline{m}, \ldots, \overline{m}, m_{K+1}, \ldots, m_N])$, where $\overline{m} = \mathcal{M}_K([m_1, \ldots, m_K])$. This property ensures the additivity of the aggregator (and is the reason behind the *quasi-arithmetic* moniker). It implies common properties in the study of inequality and mobility such as replication invariance, so replicating the population $n \in \mathbb{N}$ times does not change aggregate mobility, $\mathcal{M}_{nN}\left(\widehat{M}\right) = \mathcal{M}_N(M)$ for $\widehat{M} = \mathbb{1}_n \otimes M$.

We further pin down the aggregator by requiring $\mathcal{M}$ be homogeneous of degree 1. So, if mobility is re-scaled, aggregate mobility is re-scaled in the same way. The result is standard and we state it without proof (see Hardy, Littlewood, and Pólya 1952, Thm. 84).

LEMMA A1 (**Homogeneous Aggregation**). *Let $\mathcal{M}_N : \mathbb{R}^N \longrightarrow \mathbb{R}$ satisfy Theorem A1. $\mathcal{M}_N$ is homogeneous of degree 1, so that $\mathcal{M}_N(\kappa M) = \kappa\mathcal{M}_N(M)$ for all $\kappa > 0$, if and only if $\Gamma(m) = m^\gamma$ for some $\gamma \neq 0$, with the $\gamma = 0$ case corresponding to the geometric average. When $M \geq 0$ this gives the standard power mean,*

$$\mathcal{M}_N(M) = \left(\frac{1}{N}\sum_{i=1}^{N}(m_i)^\gamma\right)^{\frac{1}{\gamma}}. \tag{B.5}$$

When M includes negative values we instead get the equivalent signed power mean with $\Gamma(m) = sign(m)|m|^\gamma$:

$$\mathcal{M}_N(M) = sign\left(\frac{1}{N}\sum_{i=1}^{N} sign(m_i)|m_i|^\gamma\right) \times \left|\frac{1}{N}\sum_{i=1}^{N} sign(m_i)|m_i|^\gamma\right|^{\frac{1}{\gamma}}. \tag{B.6}$$

Average mobility corresponds naturally to $\overline{m} \equiv \frac{1}{N}\sum_{i=1}^{N} m_i$. It can be broken up in general into the contributions of upward and downward mobility:

$$\mathcal{U} \equiv \frac{1}{N}\sum_{i=1}^{N} |m_i|^{+}, \quad \text{and} \quad \mathcal{D} \equiv \frac{1}{N}\sum_{i=1}^{N} |m_i|^{-}, \quad \text{with} \quad \overline{m} = \mathcal{U} - \mathcal{D}. \tag{B.7}$$

When measuring symmetric mobility this reduces to $\overline{m} = \mathcal{U}$, as $\left|m_i^a\right|^{-} = 0$ for all i.

Symmetric mobility is equal to absolute net mobility, adjusted by either (absolute) downward or upward Lorenz mobility depending on whether overall inequality has increased or decreased.

Lemma A2 (**Average Symmetric Mobility**). *Under dynastic mobility (12), average mobility satisfies:*

$$\frac{1}{N}\sum_{i=1}^{N} m_i^a = \underbrace{\frac{1}{2}\left|\mathcal{G}\left(Y^P\right) - \mathcal{G}\left(Y^K\right)\right|}_{\text{Absolute Net Mobility}} + 2\left(\underbrace{\mathcal{D}^s\,\mathbb{1}_{\{\mathcal{G}(Y^P)\geq\mathcal{G}(Y^K)\}}}_{\text{Downward Lorenz Mobility}} + \underbrace{\mathcal{U}^s\,\mathbb{1}_{\{\mathcal{G}(Y^P)<\mathcal{G}(Y^K)\}}}_{\text{Upward Lorenz Mobility}}\right)$$

where $\mathcal{G}(Y) = 1 + \frac{1}{N} - \frac{2}{N}\sum_{i=1}^{N} L_i$ is the Gini coefficient and $\mathcal{U}^s$ and $\mathcal{D}^s$ measure upward and downward signed mobility.

The second and third terms in the decomposition in Corollary A2 isolate the changes in Lorenz ordinates that do not come from the overall change in inequality. Take, for instance, a decrease in inequality that reduces the Gini coefficient (a net increase in Lorenz ordinates) and implies positive net mobility (captured in the first term). Symmetric mobility accounts for changes in the Lorenz ordinates of dynasties with upward as well as downward mobility. Total upward mobility must be larger than downward mobility as it captures both the decrease in inequality and reshuffling from downward mobile dynasties. It is this reshuffling that the decomposition isolates, counting it twice, reflecting that downward moves necessarily correspond to an upward move for another dynasty when net mobility is positive.

When aggregating economic status, defined as in Proposition A2, we obtain the following characterization of generalized net mobility:

Lemma A3 (**Generalized Net Mobility**). *Under Lemma A1 with $\gamma = 1$, and Propositions A2 and A4, aggregate mobility has the form:*

$$\widetilde{\mathcal{M}}_N(M) \propto \widetilde{G}\left(Y^P\right) - \widetilde{G}\left(Y^K\right), \tag{B.8}$$

where $\widetilde{G}(Y) \equiv 1 + \frac{1}{N} - \frac{2}{N}\sum_{i=1}^{N}\Omega\left(r_i\right)L_i$ is akin to Weymark's (1981) Generalized-Gini coefficient. When $\Omega(r) = 1$ for all r we obtain the usual Gini coefficient $G(Y) = 1 + \frac{1}{N} - \frac{2}{N}\sum_{i=1}^{N} L_i$.

B.1. Aggregating unequal mobility

We now turn to how the curvature of the aggregator, γ, determines the emphasis on broad-based versus concentrated mobility, as in measures of inequality aversion (Atkinson 1970) or choice under uncertainty (Ackerberg, Hirano, and Shahriar 2017). When $\gamma = 1$ we recover the average mobility, which neither favors nor penalizes dispersion in mobility. For positive γ, lower values favor broad-based mobility while higher values favor dispersion, effectively placing a higher weight on dynasties with the highest mobility. Negative values of γ place more weight on the dynasties with the lowest mobility.

We can better see the role of γ by expressing aggregate mobility as *average mobility* corrected by *mobility concentration*, as in Atkinson (1970). This is immediate when mobility is positive, in which case aggregate mobility is

$$\mathcal{M}_N(M) = \underbrace{\overline{m}}_{\text{Average Mobility}} \times \overbrace{\exp\left(\frac{\gamma-1}{\gamma} D_\gamma(M)\right)}^{\text{Mobility Concentration}}, \tag{B.9}$$

where average mobility is $\overline{m} \equiv \frac{1}{N}\sum_{i=1}^{N} m_i$ and mobility concentration corresponds to the Rényi divergence of the relative upward mobility shares $r_i \equiv \frac{m_i}{N\overline{m}} \geq 0$: $D_\gamma(M) = \log N - H_\gamma(r)$, with $H_\gamma(r) = \frac{1}{1-\gamma}\log\left(\sum_{i=1}^{N}(r_i)^\gamma\right)$ the corresponding Rényi entropy.[3] More generally, mobility can be either positive or negative. These cases need to be addressed separately, as we show in the Appendix.

The first term in (B.9) aggregates mobility linearly and provides a direct link between changes in inequality and mobility as shown in Lemmas 2 and A2. The second term in (B.9) captures how concentrated mobility is between dynasties. The weight on mobility concentration depends naturally on the value of γ. When $\gamma = 1$, concentration does not matter. Lower values of γ penalize concentration, placing more weight on lower values of mobility, and higher values of γ favor concentration, placing more weight on higher values of mobility.

In addition, when $\gamma \neq 1$, the net mobility concentration term in equation (B.9) can be interpreted as the equivalent fall in inequality required to exactly offset unequal mobility. This completes the link between mobility and inequality even when the Lorenz curves of Y^K and Y^P intersect.

In general, we measure mobility concentration, $\mathcal{C}$, separately within upward and downward mobility. Upward mobility concentration is

$$\mathcal{C}^+_\gamma(M) \equiv \exp\left(\frac{\gamma-1}{\gamma} D^+_\gamma(M)\right), \tag{B.10}$$

[3]The Rényi entropy is a general measure of dispersion allowing for curvature. It builds on the use of information-theoretic concentration measures for inequality (Theil 1967) and follows the equally-distributed-equivalent tradition of Kolm (1969), Atkinson (1970), and Sen (1973).

where $D^+_\gamma(M) = \log N - H^+_\gamma(r)$ is the Rényi divergence of the relative upward mobility shares $|r_i|^+ \equiv \frac{|m_i|^+}{N\mathcal{U}} \geq 0$. Their Rényi entropy is $H^+_\gamma(r) = \frac{1}{1-\gamma}\log\left(\sum_{i=1}^N \left(|r_i|^+\right)^\gamma\right)$. This is a general measure of dispersion allowing for curvature.

The following lemma formalizes the roles of average mobility and mobility concentration.

Lemma A4 (**Average mobility and mobility concentration**). *Let* $\overline{m} \equiv \frac{1}{N}\sum_{i=1}^N m_i \neq 0$. *Aggregate mobility under* (B.5) *satisfies:*

$$\mathcal{M}_N(M) = \underbrace{\overline{m}}_{\textit{Average Mobility}} \times \overbrace{\frac{sign(\mathcal{S}_\gamma)}{sign(\overline{m})}}^{\gamma\textit{-sign correction}} \times \underbrace{\overline{\mathcal{C}}(M)}_{\textit{Net Mobility Concentration}}, \tag{B.11}$$

where $\mathcal{S}_\gamma(M) \equiv \sum_{i=1}^N sign(m_i)|m_i|^\gamma$ *is the signed aggregate of mobility as in Lemma A1 and*

$$\overline{\mathcal{C}}(M) \equiv \left|\left(\frac{\mathcal{U}}{|\overline{m}|}\right)^\gamma \left(\mathcal{C}^+_\gamma(M)\right)^\gamma - \left(\frac{\mathcal{D}}{|\overline{m}|}\right)^\gamma \left(\mathcal{C}^-_\gamma(M)\right)^\gamma\right|^{\frac{1}{\gamma}} \tag{B.12}$$

is net mobility concentration, with upward and downward mobility as in (B.7) *and upward and downward mobility concentration as in* (B.10).

Proof. The result follows from manipulation of the aggregator in (B.5). □

This collapses to the product of average mobility and overall mobility concentration when $M \geq 0$:

$$\mathcal{M}_N(M) = \underbrace{\overline{m}}_{\text{Average Mobility}} \times \underbrace{\exp\left(\frac{\gamma-1}{\gamma}D_\gamma(M)\right)}_{\text{Mobility Concentration}}, \tag{B.13}$$

When mobility is signed, the middle term corrects the sign of aggregate mobility as the curvature γ can make $\mathcal{S}_\gamma$ have a different sign from $N\overline{m} = \mathcal{S}_1$.

B.2. Bounds on symmetric mobility

We consider the maximum mobility, $\mathcal{M}$, when $\gamma = 1$ given two income distributions. This corresponds to the value of an assignment from generation P to generation K. Order dynasties in Y^G such that $y^G_i \leq y^G_{i+1}$ for all $i = 1, \ldots, N$ and $G \in \{P, K\}$. The index i determines the order of income and not the identity of the dynasty. Denote by s^G_i the corresponding status of the dynasty with income position i in generation G.

We know that $s_i^G \geq 0$ is increasing in i. The assignment (or transport) that maximizes mobility solves

$$\max_T \frac{1}{N} \sum_{i=1}^{N} \left| s_{T(i)}^K - s_i^P \right| \tag{B.14}$$

The optimal assignment is counter-monotone because pair-wise mobility is represented by a Monge matrix M (a sub-modular discrete function) with typical entry $m_{ij} = \left| s_j^K - s_i^P \right|$, so that $T^\star(i) = N + 1 - i$. This result is a direct application of the rearrangement of linear sums in Burkard, Dell'Amico, and Martello (2012, Prop. 5.7). The key step requires establishing that M satisfies the Monge property, that for $1 \leq i < k \leq n$ and $1 \leq j < \ell \leq n$ the entries of M satisfy

$$m_{ij} + m_{k\ell} \leq m_{i\ell} + m_{kj} ; \tag{B.15}$$

$$\left| s_j^K - s_i^P \right| + \left| s_\ell^K - s_k^P \right| \leq \left| s_\ell^K - s_i^P \right| + \left| s_j^K - s_k^P \right| . \tag{B.16}$$

To prove this, consider $i < k$ and define

$$f(x) \equiv \left| x - s_i^P \right| - \left| x - s_k^P \right| = \begin{cases} s_i^P - s_k^P, & x \leq s_i^P , \\ 2x - (s_i^P + s_k^P), & s_i^P \leq x \leq s_k^P , \\ s_k^P - s_i^P, & x \geq s_k^P . \end{cases} \tag{B.17}$$

Hence, f is non-decreasing in x. We know $s_j^K \leq s_\ell^K$ because $j < \ell$, so $f\left(s_j^K\right) \leq f\left(s_\ell^K\right)$:

$$\left| s_j^K - s_i^P \right| - \left| s_j^K - s_k^P \right| \leq \left| s_\ell^K - s_i^P \right| - \left| s_\ell^K - s_k^P \right| . \tag{B.18}$$

Rearranging yields the inequality in (B.16), following from the sub-modularity of m.

To prove that the counter-monotone assignment is optimal, consider any candidate assignment T that is not strictly decreasing, so there is $i < k$ with $T(i) < T(k)$. Swap the assignment of i and k, leading to a new assignment with $\widehat{T}(i) = T(k)$, $\widehat{T}(k) = T(i)$, and $\widehat{T}(\omega) = T(j)$ for $\omega \neq i, k$. Then, set $j = T(i)$ and $\ell = T(k)$ and apply (B.16) to get that

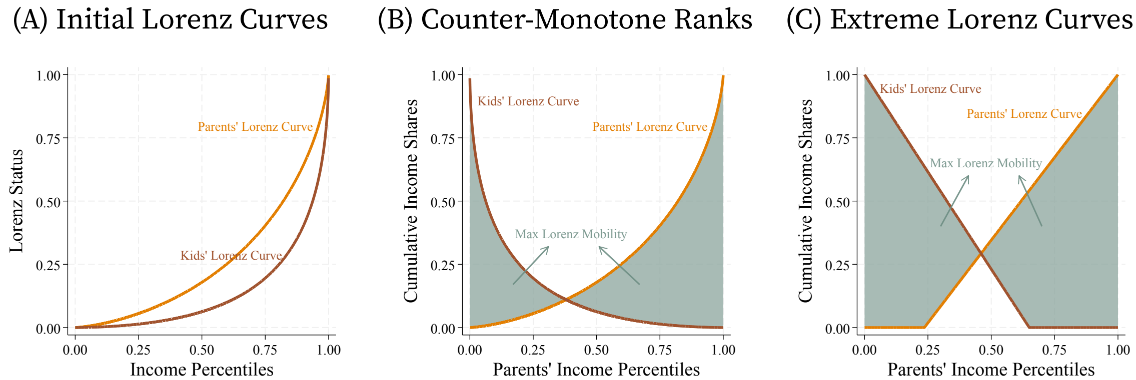
$$m_{i,T(i)} + m_{k,T(k)} \leq m_{i,\widehat{T}(i)} + m_{k,\widehat{T}(k)} . \tag{B.19}$$

So, the new assignment weakly improves mobility. Save from ties in status, only the counter-monotone assignment is strictly decreasing and maximizes mobility.

Horizontal mobility is also maximized by the counter-monotone assignment as it consists of the special case when the two income distributions coincide and $s_i^K = s_i^P$.

The upper envelope of mobility is obtained by maximising over Lorenz curves.

FIGURE B.1. Maximal Lorenz Mobility



Notes: Panel A shows the parental and child Lorenz curves. Holding these curves fixed, Panel B pairs parental and child ranks counter-monotonically, which maximizes average absolute Lorenz mobility. Panel C shows the extremal curves used to maximize mobility over Lorenz curves; they have the form $\tilde{L}_c(r) = {}^{r-c}/_{1-c}$ for $r > c$ and 0 otherwise. The shaded areas represent aggregate symmetric mobility under $\gamma = 1$.

For analytic convenience, we compute this in the limiting case as $N \longrightarrow \infty$.[4] Baillo, Carcamo, and Mora-Corral (2022) show that the extreme points in Lorenz space are piecewise linear functions. As the objective function is convex in the Lorenz curves and we maximise over a convex set, it is sufficient to search for the kink in the two-segment Lorenz curves maximising mobility.[5] That is Lorenz curves defined by a kink $c^G \in [0, 1]$ so that $\tilde{L}_{c^G}(r) = 0$ for $r \in [0, c]$, and $\tilde{L}_{c^G}(r) = \frac{r-c^G}{1-c^G}$ for $r \in (c, 1]$.

Mobility is therefore bounded by the maximum under the counter-monotone assignment, obtained by integrating $\tilde{m}(r) = \left| \tilde{L}_{c^K}(1-r) - \tilde{L}_{c^P}(r) \right|$,

$$\max_{c^P, c^K \in [0,1]} \int_0^1 \left| \tilde{L}_{c^K}(1-r) - \tilde{L}_{c^P}(r) \right| dr \, . \tag{B.20}$$

There are two cases determined by the breaks in (B.20) at $r = c^P$ and $r = 1 - c^K$:

(a) $0 \leq c^P < 1 - c^K \leq 1$: There are four intervals to consider. One curve is zero in two intervals; the other two intervals are determined by the curves' crossing at

[4] The counter-monotone assignment can now be expressed in terms of ranks, so that $T^\star(r) = 1 - r$. The argument is the same as above but builds directly on the sub-modularity of the difference in status, now defined as functions of rank, r.

[5] Formally, Baillo, Carcamo, and Mora-Corral show that this is the extremal curve for any given value of the Gini coefficient. Searching over kink-points then amounts to searching over Gini coefficients.

$r = \frac{1-c^K}{2-c^P-c^K}$.

$$\tilde{m}(r) = \begin{cases} \frac{1-c^K-r}{1-c^K} & r \in \left[0, c^P\right] ; \\ \frac{1-c^K-r}{1-c^K} - \frac{r-c^P}{1-c^P} & r \in \left(c^P, \frac{1-c^K}{2-c^P-c^K}\right] ; \\ \frac{r-c^P}{1-c^P} - \frac{1-c^K-r}{1-c^K} & r \in \left(\frac{1-c^K}{2-c^P-c^K}, 1-c^K\right) ; \\ \frac{r-c^P}{1-c^P} & r \in \left[1-c^K, 1\right] . \end{cases} \tag{B.21}$$

The integral is then

$$\int_0^1 \tilde{m}(r)\, dr = \frac{1+c^P}{2} - \frac{\left(c^P\right)^2}{2\left(1-c^K\right)} + \frac{\left(1-c^P-c^K\right)^2\left(\left(1-c^P\right)^2 + \left(1-c^K\right)^2\right)}{2\left(1-c^K\right)\left(1-c^P\right)\left(2-c^P-c^K\right)} - \frac{\left(1-c^P-c^K\right)^2}{2\left(1-c^P\right)} . \tag{B.22}$$

(b) $0 \le 1-c^K \le c^P \le 1$: There is a gap between $1-c^K$ and c^P where both functions are zero. So the integral is

$$\int_0^1 \tilde{m}(r)\, dr = \int_0^{1-c^K} 1 - \frac{r}{1-c^K}\, dr + \int_{c^P}^1 \frac{r-c^P}{1-c^P}\, dr = \frac{1-c^K}{2} + \frac{1-c^P}{2} . \tag{B.23}$$

When $1-c^K = c^P$ the lines touch but do not intersect and the integral is equal to 1/2. We can now establish the maximal mobility by selecting the values of c^P and c^K. Crucially, mobility does not depend on the values of c^P and c^K independently, but on the gap between these values and 1: $g \equiv 1-c^P-c^K$. The first case above corresponds to $g > 0$ and the second case to $g \le 0$. Then, mobility is

$$\int_0^1 \tilde{m}(r)\, dr = \begin{cases} \frac{1+2g-g^2}{2(1+g)} & g > 0 ; \\ \frac{1+g}{2} & g \le 0 . \end{cases} \tag{B.24}$$

The second case is maximized when $g = 0$ and mobility is 1/2. The critical values of the first case are the roots of $1-2g-g^2$. The only admissible root is $g^\star = \sqrt{2}-1$, which delivers a value of $2-\sqrt{2} \approx 0.59$.

Therefore, the maximal mobility is attained by Lorenz curves of the form $\tilde{L}_{c^G}$ under counter-monotone assignments between generations such that $c^P + c^K = 2-\sqrt{2}$. This tight upper bound implies that

$$0 \le \mathcal{M}(M) \le 2-\sqrt{2} . \tag{B.25}$$

Moreover, this is also the upper bound for horizontal mobility, equivalent to constraining $c^P = c^K = c$ and setting $c = 1-\frac{1}{\sqrt{2}}$.

B.3. Progressive taxation

We illustrate the implications of inequality for average mobility by discussing how it responds to taxation. For this, consider the case where market income in each generation is Pareto-distributed with tail parameter $\alpha^G > 1$, $G \in \{P, K\}$, so that the Lorenz curve and the Gini coefficient are

$$\widetilde{L}^G(r) = 1 - (1-r)^{1-\frac{1}{\alpha^G}}, \qquad \text{and} \qquad \mathcal{G}\left(Y^G\right) = \frac{1}{2\alpha_G - 1}. \tag{B.26}$$

We further assume disposable income, $\widehat{y}$, satisfies $\widehat{y}^G = \lambda^G y^{1-\tau^G}$ as in Feldstein (1969); Bénabou (2002); Heathcote, Storesletten, and Violante (2014), where $\lambda^G > 0$ captures the average tax rate and $\tau^G \in [0, 1)$ controls the progressivity, or how quickly average tax rates rise with income. In this case, disposable income is also Pareto-distributed with a tail parameter $\alpha^G/(1 - \tau^G)$.

The irrelevance of taxation for ranks extends to any rank-based dynastic mobility measure $m(r^P, r^K)$. This includes measures of rank mobility commonly reported such as the intergenerational rank correlation, corr $\left(r_i^P, r_i^K\right)$, conditional expected rank for children born to parents at the p$^{\text{th}}$ percentile, $E[r_i^K | r_i^P = p]$, or the probability a child born to parents in the bottom-quintile is in the top-quintile of their own generation's distribution (or any other transition between segments of the distribution).

The effect of taxation on mobility is immediate in the case of net mobility where mobility is given by the change in the Gini coefficient (Corollary 2). A more progressive tax system—higher τ—makes the tail thinner as it reduces inequality. Net mobility reflects changes in inequality and progressivity so that it is positive when $\frac{\alpha^K}{\alpha^P} > \frac{1-\tau^K}{1-\tau^P}$. In general, an increase in progressivity in the children's generation increases net mobility while more progressivity in the parents' generation decreases it. Moreover, tax progressivity affects mobility even when it is constant over time by attenuating differences in income that result in an attenuation of overall mobility.[6]

Lemma A5 (**Progressive Taxation and Net Mobility**). *Under Corollary 2, assume Pareto-distributed income in each generation $G \in \{P, K\}$, with tail parameter $\alpha^G > 1$, and disposable income satisfying the log-linear tax schedule, with progressivity parameter $\tau^G \in [0, 1)$. Then, before-tax and after-tax net mobility satisfy*

(a) *Before-tax net mobility is positive if and only if $\frac{\alpha^K}{\alpha^P} > 1$ and after-tax net mobility is positive if and only if $\frac{\alpha^K}{\alpha^P} > \frac{1-\tau^K}{1-\tau^P}$.*

(b) *Increasing progressivity in the children's generation increases net mobility, while increasing progressivity in the parents' generation decreases it.*

[6]The effect of progressivity on Lorenz mobility differs from the effect on measures of income-mobility like the intergenerational income elasticity (IGE). The IGE depends only on relative progressivity across generations so that common progressivity leaves the IGE unchanged. This is likely to occur in applications where the income of both generations is measured in the same period. See Appendix B.3.

(c) *If progressivity is common across generations, $\tau^P = \tau^K = \tau$, then taxation preserves the direction of net mobility and attenuates its magnitude toward zero.*

PROOF. From Corollary 2, net mobility before and after taxes satisfy, respectively,

$$\mathcal{M}_N\left(M^s\right) \propto \frac{1}{2\alpha^P - 1} - \frac{1}{2\alpha^K - 1} = \frac{2\left(\alpha^K - \alpha^P\right)}{\left(2\alpha^P - 1\right)\left(2\alpha^K - 1\right)}, \tag{B.27}$$

$$\widehat{\mathcal{M}}_N\left(M^s\right) \propto \frac{1-\tau^P}{2\alpha^P + \tau^P - 1} - \frac{1-\tau^K}{2\alpha^K + \tau^K - 1}. \tag{B.28}$$

These expressions immediately give $\mathcal{M}_N\left(M^s\right) > 0$ if and only if $\frac{\alpha^K}{\alpha^P} > 1$, and $\widehat{\mathcal{M}}_N\left(M^s\right) > 0$ if and only if $\frac{\alpha^K}{\alpha^P} > \frac{1-\tau^K}{1-\tau^P}$. Taking derivatives gives the response of after-tax mobility to tax progressivity in each generation:

$$\frac{\partial\widehat{\mathcal{M}}_N\left(M^s\right)}{\partial\tau^K} = \frac{\alpha^K}{\left(2\alpha^K + \tau^K - 1\right)^2} > 0 \quad \text{and} \quad \frac{\partial\widehat{\mathcal{M}}_N\left(M^s\right)}{\partial\tau^P} = -\frac{\alpha^P}{\left(2\alpha^P + \tau^P - 1\right)^2} < 0\,. \tag{B.29}$$

Finally, setting $\tau^P = \tau^K = \tau > 0$ we obtain

$$\widehat{\mathcal{M}}_N\left(M^s\right) \propto \frac{2\left(1-\tau\right)\left(\alpha^K - \alpha^P\right)}{\left(2\alpha^P + \tau - 1\right)\left(2\alpha^K + \tau - 1\right)}, \tag{B.30}$$

$$\frac{\left|\widehat{\mathcal{M}}_N\left(M^s\right)\right|}{\left|\mathcal{M}_N\left(M^s\right)\right|} = \frac{\left(1-\tau\right)\left(2\alpha^P - 1\right)\left(2\alpha^K - 1\right)}{\left(2\alpha^P + \tau - 1\right)\left(2\alpha^K + \tau - 1\right)} < 1 \tag{B.31}$$

□

The role of progressive taxation in other common mobility measures is stark. Consider the intergenerational elasticities of income and disposable income discussed in footnote 6, respectively,

$$\log\left(y_i^K\right) = \beta_0 + \beta_1 \log\left(y_i^P\right) + u_i \tag{B.32}$$

$$\log\left(y_{i,\text{disp}}^K\right) = \beta_0^{\text{disp}} + \beta_1^{\text{disp}} \log\left(y_{i,\text{disp}}^P\right) + u_{i,\text{disp}}, \tag{B.33}$$

which, under log-linear taxation, yield

$$\beta_1^{\text{disp}} = \beta_1 \frac{\left(1-\tau_K\right)}{\left(1-\tau_P\right)}. \tag{B.34}$$

Thus, the intergenerational elasticity of disposable income differs from that of market income only by the relative difference in progressivity. When both generations face the same progressive tax system, for example if market income of both generations is measured at the same point in time (as in many data sources that observe different ages at the same date), then the intergenerational elasticity also inherits an irrelevance of progressive taxation.

The following lemma makes the connection between rank dependence and progressive taxation explicit by combining Pareto-distributed income as in (B.26) with the Farlie-Gumbel-Morgenstern (FGM) copula describing the correspondence of ranks between generations. The usefulness of the FGM copula is that it gives a closed-form single-parameter condition on the dependence parameter. We state the result below:

LEMMA A6 (**Progressive Taxation and Symmetric Mobility with FGM Dependence**). *Suppose incomes in generation $G \in \{K, P\}$ are Pareto with parameter $\alpha_G > 1$ and tax progressivity $\tau_G \in [0, 1)$. Let*

$$p_G \equiv \frac{\alpha_G}{\alpha_G - 1 + \tau_G}, \tag{B.35}$$

denote the curvature of the after-tax Lorenz curve. If the joint distribution of incomes is described by the FGM copula,

$$C_\theta(u, v) = uv\left[1 + \theta(1 - u)(1 - v)\right], \qquad \theta \in [-1, 1], \tag{B.36}$$

then, under aggregator (B.5) with $\gamma = 1$ and dynastic mobility (12),

$$\mathcal{M}_N(M^a) \propto \frac{p_K}{p_K + 1} + \frac{p_P}{p_P + 1} - 2\int_0^1 C_\theta\left(1 - t^{p_K}, 1 - t^{p_P}\right) dt, \tag{B.37}$$

where the effect of (common-)tax progressivity is affine in θ and there exists a positive threshold θ_G^* such that progressivity raises symmetric mobility if and only if $\theta < \theta_G^*$.

Appendix C. Comparing Lorenz and Rank Mobility in Matched Data: Norwegian Administrative Records

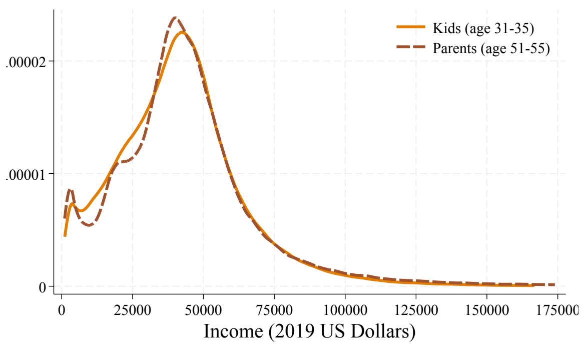
We now apply our framework to study the intergenerational mobility of Norwegians so as to establish how Lorenz mobility compares with rank mobility. We find Lorenz mobility is lower than rank mobility; compared to their respective maximal levels, Lorenz mobility is over 10 percentage points lower than rank mobility.

Norwegian Data. We use detailed longitudinal data on individual income collected by the Norwegian tax authority between 1993 and 2017 as well as demographic information available in its population files. We focus on individual market income from wages and capital. See Blundell, Graber, and Mogstad (2015) for details on the Norwegian income tax records.

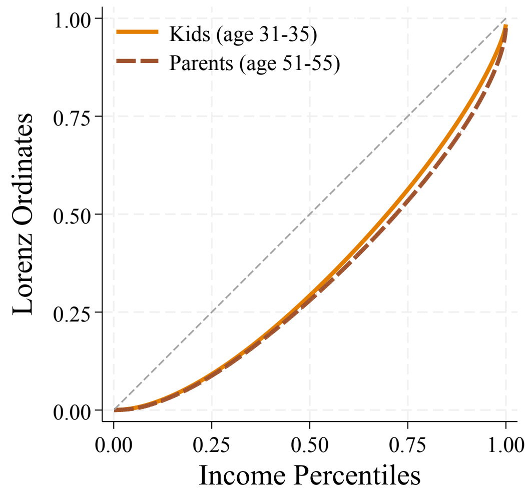
There are several key dimensions in which the Norwegian administrative data provide us with an appropriate benchmark for studying Lorenz and rank mobility. The data covers the entire population, including the richest Norwegians, allowing us to

FIGURE C.1. Income distribution across generations

(A) Density of income - Norway



(B) Lorenz curves - Norway



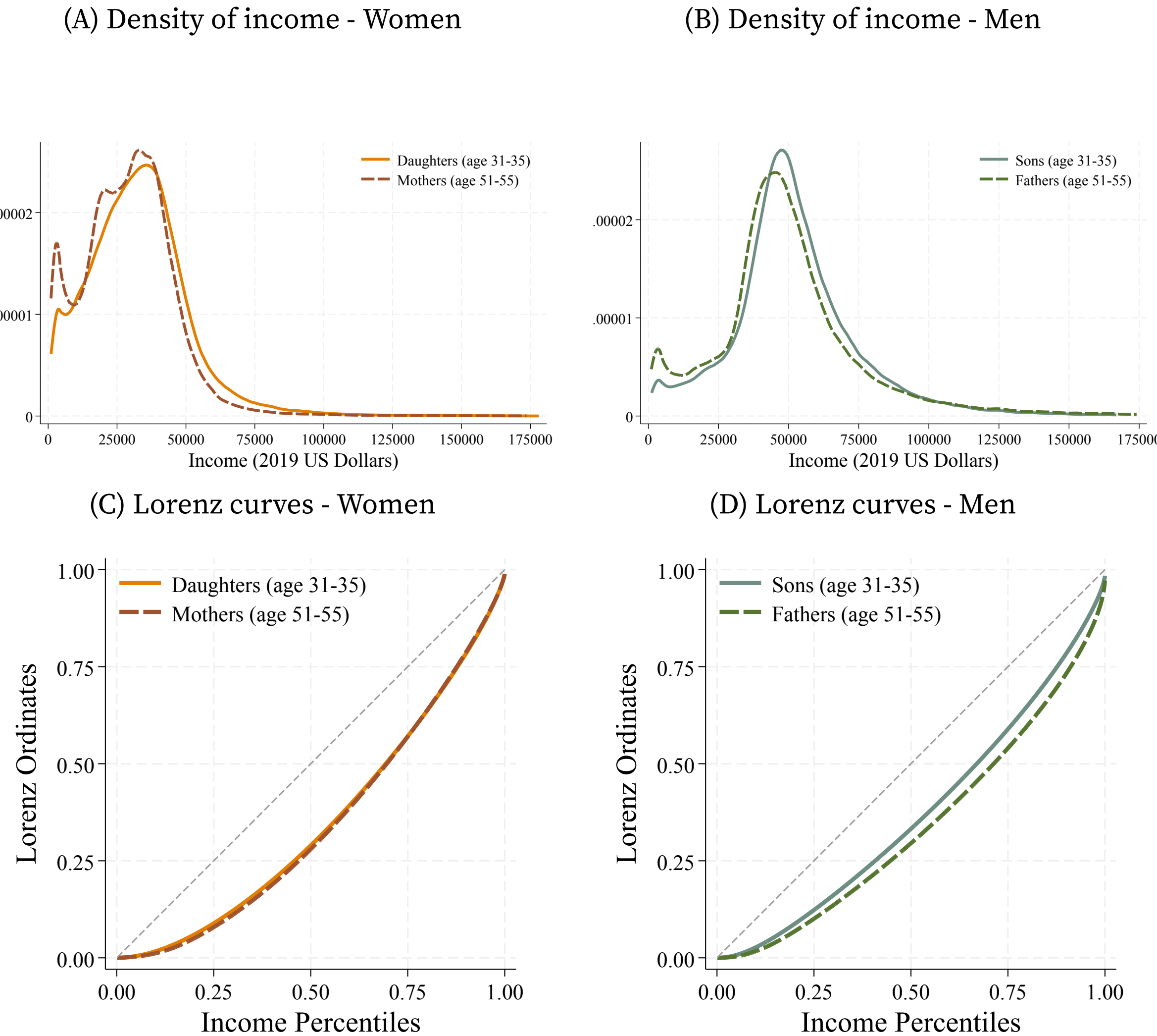
Notes: Panel A shows kernel density estimates of income for children and parents. Panel B shows the Lorenz curves for the corresponding income distributions.

construct accurate income ranks and Lorenz ordinates. Second, income is third-party reported, eliminating concerns about measurement error from self-reporting and censoring. Third, we link incomes across generations without relying on imputation (e.g., Collins and Wanamaker 2022; Ward 2023; or Jácome, Kuziemko, and Naidu 2025) or bounding exercises (e.g., Chetty et al. 2017; Berman 2022; or Manduca et al. 2024). Finally, the Norwegian data allows us to revisit the relationship between mobility in market and disposable income. We take advantage of the fact that the data are collected by the Norwegian Tax Authority (*Skatteetaten*), which provides accurate measures of levied taxes at the individual level and allows us to capture effective taxation. Redistribution through taxes and transfers adds to the decrease in inequality already present in market income.

We study Norwegians born between 1961 and 1970 and record their average income between the ages of 31 and 35. We drop individuals who earn less than $1,000 USD before taxes and transfers on average (all dollar amounts are measured in 2019 dollars). We link these Norwegians to their fathers and mothers and record the maximum of their parents' average income between the ages of 51 and 55. This leaves us with 303,451 paired observations.

We construct two additional samples with the Norwegian data in which we distinguish between female and male mobility. First, we construct a sample of mothers and daughters, defined by female Norwegians born between 1961 and 1970 with recorded information on their mothers. This sample contains a total of 128,235 paired observations. Second, we construct an equivalent sample for sons and their fathers, with a total of 101,529 observations.

FIGURE C.2. Income distribution across generations in Norway for women and men



Notes: Panels A and B show kernel density estimates of individual market income for daughters and mothers and for sons and fathers, respectively. Panels C and D show the Lorenz curves for the corresponding income distributions. Income is averaged over ages 31–35 for daughters and sons and ages 51–55 for mothers and fathers and is measured in 2019 US dollars.

The income distribution of Norwegian children is shifted to the left relative to that of their parents. Children's average income is 7.6% lower than their parents' average income. The parental distribution also has less mass concentrated at low incomes (except at the very bottom) and more mass at incomes above 100,000 dollars (Figure C.1A). Nevertheless, these differences in income translate into small but visible differences in the Lorenz curves between generations (Figure C.1B). The Lorenz curve of children's income is above that of parental income throughout the distribution, evidencing a lower degree of inequality. These differences become more marked above the median.

We consider signed and symmetric mobility (as in equations 11 and 12) aggregated using the homogeneous aggregator in (B.5) under different curvatures. Table C.1 presents the results. We also examine standard measures of intergenerational correlations for log-income, ranks, and Lorenz ordinates (as in 20). We present these in Appendix C.

Signed mobility. Average signed mobility ($\gamma = 1$) reflects the changes in inequality between generations. The Gini falls by almost 3.2 percentage points for market income and by 6.2 percentage points for disposable income, nearly twice as much. Ranks capture only horizontal mobility and so their aggregate change is 0 by construction.

Mobility itself is also unequal. We see this by comparing our main measure of aggregate mobility $\mathcal{M}$ without curvature, $\gamma = 1$, to measures with lower and higher curvature. When we set $\gamma = 1/2$, aggregate mobility favors a more even distribution of mobility, reducing $\mathcal{M}$ if mobility is concentrated among a few dynasties. Indeed, Lorenz mobility is reduced to essentially zero. Rank mobility is similarly concentrated and remains negligible with $\gamma = 1/2$.

Conversely, $\mathcal{M}$ places more weight on dynasties with higher mobility when setting $\gamma = 2$. For Norway, Lorenz and rank mobility go in different directions. While positional mobility is dominated by large downward moves (3.8 percentage points on average), the decrease in inequality gives large upward Lorenz mobility (over 9 percentage points on average). This is consistent with Norway's more compressed income distribution, which results in more rank displacement that nevertheless has little impact on mobility. These patterns are also present in disposable income.

Symmetric mobility. We find symmetric Lorenz mobility is markedly lower than rank mobility, reflecting the role of income inequality in shaping differences across generations. Measured relative to their maximal values, Lorenz mobility is over 10 percentage points lower than rank mobility. Average Lorenz mobility is 0.28, or 47% of its maximal value, reflecting Norway's decrease in inequality across generations. Rank mobility is 0.3, or 60% of its maximal value.

In this case, both Lorenz and rank mobility primarily reflect horizontal mobility (as the Lorenz curves of the two generations are relatively close together), but weight positional moves differently, reflecting the role of income inequality. Unlike signed mobility, symmetric Lorenz mobility is largely unaffected by taxation. Thus, on net, the tax system produces additional upward mobility without materially changing the overall movement in Lorenz ordinates. Rank mobility is essentially unchanged. Minor differences arise because transfers can break the monotonicity of the tax system, so ranks under market and disposable income need not coincide exactly.

Finally, symmetric mobility is also concentrated, but not as much as signed mobility. Penalizing concentration (setting $\gamma = 1/2$) reduces Lorenz and rank mobility by 16–18%. Placing more weight on high-mobility dynasties (setting $\gamma = 2$) increases measured rank mobility by 26% and Lorenz mobility by close to 30%. Moreover, Lorenz mobility remains below rank mobility under all three values of γ, so accounting for mobility

TABLE C.1. Aggregate mobility in Norway

		Aggregate Mobility: $\mathcal{M} = \left(\frac{1}{N}\sum(m_i)^{\gamma}\right)^{\frac{1}{\gamma}}$					
		Signed Mobility			Symmetric Mobility		
		$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$	$\gamma = 1$	$\gamma = 1/2$	$\gamma = 2$
Market Income							
Norway	Lorenz	0.016	0.001	0.092	0.275	0.227	0.351
	Ranks	—	0	–0.038	0.299	0.251	0.375
Women	Lorenz	0.005	0.000	0.053	0.285	0.234	0.365
	Ranks	—	0	0.008	0.297	0.249	0.372
Men	Lorenz	0.033	0.003	0.128	0.261	0.214	0.334
	Ranks	—	0	–0.041	0.283	0.235	0.360
Disposable Income							
Norway	Lorenz	0.031	0.002	0.129	0.277	0.232	0.350
	Ranks	—	0	–0.036	0.298	0.250	0.374
Women	Lorenz	0.018	0.001	0.100	0.291	0.243	0.367
	Ranks	—	0	0.020	0.298	0.250	0.372
Men	Lorenz	0.027	0.002	0.113	0.262	0.218	0.333
	Ranks	—	0	–0.052	0.286	0.238	0.362

Notes: The table reports aggregate signed and symmetric mobility in Lorenz ordinates and income ranks for the Norwegian samples. Women denotes mother–daughter pairs, Men denotes father–son pairs, and Norway denotes the pooled parent–child sample. The upper panel uses market income and the lower panel uses disposable income. Signed mobility is the change in a dynasty's status, while symmetric mobility is the magnitude of that change. Aggregate signed mobility is computed using the signed power mean in Lemma A1; aggregate symmetric mobility uses the standard power mean. The case $\gamma = 1$ is the arithmetic average, $\gamma = 1/2$ places relatively more weight on broadly shared mobility, and $\gamma = 2$ places relatively more weight on larger movements. The signed-rank entries for $\gamma = 1$ are omitted because average signed rank mobility is zero by construction.

concentration does not alter their ordering.

Women's and men's mobility in Norway. We also reproduce our analysis for Norwegian women and men, measuring mobility between mothers and daughters and fathers and sons separately. We present these results in Appendix C. Our results for mothers and daughters echo our previous findings: measured mobility is almost entirely horizontal, as inequality stayed roughly constant across generations despite significant income growth of 18% between generations. Income inequality decreased between mothers and daughters by 0.5 percentage points (a similar magnitude to the increase in inequality in the US). In contrast, we find an important role for decreased inequality among men in driving vertical mobility, with a decrease of 6.6 percentage points in the Gini coefficient. Nevertheless, there is more horizontal mobility among women, reflected in larger symmetric mobility than among men. In both cases we verify that Lorenz mobility is lower than rank mobility.

The income distributions of daughters and sons move rightwards relative to their mothers and fathers, reflecting overall income growth, and have less mass concentrated at low incomes, as seen in Figures C.2A and C.2B. Daughters' income is 18% higher than their mothers', with the bottom 10 percentiles of the distribution growing more than 70%. Sons' income growth is smaller, slightly below 11%, and less widespread, with the top 15 percentiles of sons actually mapping to lower income levels than their fathers'. After-tax income is markedly less unequal.

Changes in income distributions across generations are reflected in the Lorenz curves. As seen in Figure C.2C, there are no significant differences between the Lorenz curves of daughters and mothers, despite crossing at the top of the income distribution. In contrast, the Lorenz curve of sons shifts towards equality (Figure C.2D).

Intergenerational mobility. We find higher mobility in economic status for women than for men across most mobility measures presented in Table C.1. As we see from our main mobility measure ($\mathcal{M}$), women move on average by 29% of their aggregate income compared with men's 26%. Mobility among women corresponds mostly to horizontal mobility, reflecting changes in the position of dynasties, as there are only small differences between the Lorenz curves of mothers and daughters (Figure C.2C). The lack of vertical mobility is further evidenced by low net mobility ($\widetilde{\mathcal{M}}$).

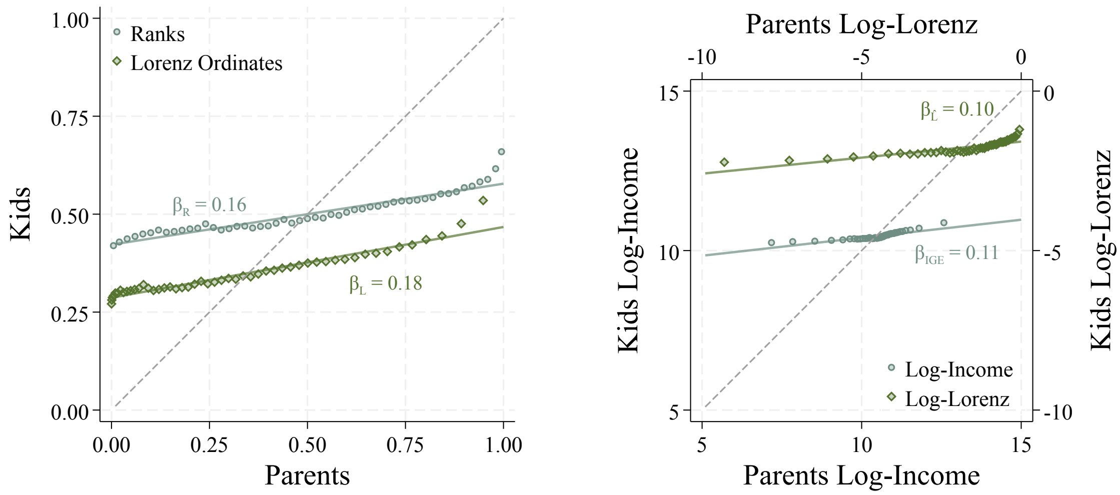
Mobility is broad-based for women and men alike. When we set $\gamma = 1/2$, aggregate mobility favors a more even distribution of mobility, reducing $\mathcal{M}$ if mobility is concentrated among a few dynasties. However, mobility remains high, with women moving by 23% of total income and men moving by 22%. Conversely, $\mathcal{M}$ increases when $\gamma = 2$, as it places more weight on dynasties with higher mobility. These results imply mobility is broad-based, but some dynasties experience larger changes.

Intergenerational correlations. Finally, we examine standard measures of intergenerational correlations for log-income, ranks, and status levels. We report these in Table C.2 and the corresponding scatter plots in Figure C.3. Outcomes are more strongly correlated across generations in Norway under every measure, consistent with the lower levels of aggregate mobility in Table C.1. The correlation of Lorenz

Figure C.3. Intergenerational correlations in Norway

(A) Lorenz and Rank Correlations

(B) Inter-Generational Elasticities



Notes: Panel A plots children's income ranks (circles) and Lorenz ordinates (diamonds) against the corresponding parental values. Panel B plots children's log-income and log-Lorenz ordinates, using separate axes for the two measures. Points are binned means, solid lines are linear fits, and dashed lines show equality across generations. The reported β coefficients are the slopes of the fitted lines.

ordinates is 0.180, while the corresponding rank correlation is 0.156. The same pattern applies when comparing the intergenerational elasticity of income and the correlation of log-Lorenz ordinates. Lorenz ordinates are more correlated across generations than ranks, by 0.024, again consistent with our previous result that Lorenz mobility is lower than rank mobility.[7] As before, this reflects churn in ranks between dynasties with similar incomes: these positional changes can be sizable in rank units even though they produce small changes in Lorenz ordinates.

[7]This corresponds to one-third of the regional differences in Bratberg et al. (2017) for Norway.

TABLE C.2. Intergenerational correlations in Norway

	Intergenerational Correlations $\rho\left(x_i^P, x_i^K\right)$			
	Lorenz	Rank	Log-Lorenz	Log-income
Market Income				
Norway	0.180	0.156	0.100	0.113
Women	0.181	0.168	0.109	0.108
Men	0.254	0.223	0.124	0.121
Disposable Income				
Norway	0.185	0.163	0.116	0.092
Women	0.174	0.168	0.098	0.102
Men	0.243	0.212	0.181	0.137

Notes: The table reports Pearson correlations between parental and child outcomes in the Norwegian samples. Women denotes mother–daughter pairs, Men denotes father–son pairs, and Norway denotes the pooled parent–child sample. The upper panel uses market income and the lower panel uses disposable income. The Lorenz and Rank columns use Lorenz ordinates and income ranks in levels; the Log-Lorenz and Log-income columns use their logarithms. Higher correlations indicate greater intergenerational persistence and therefore lower mobility.

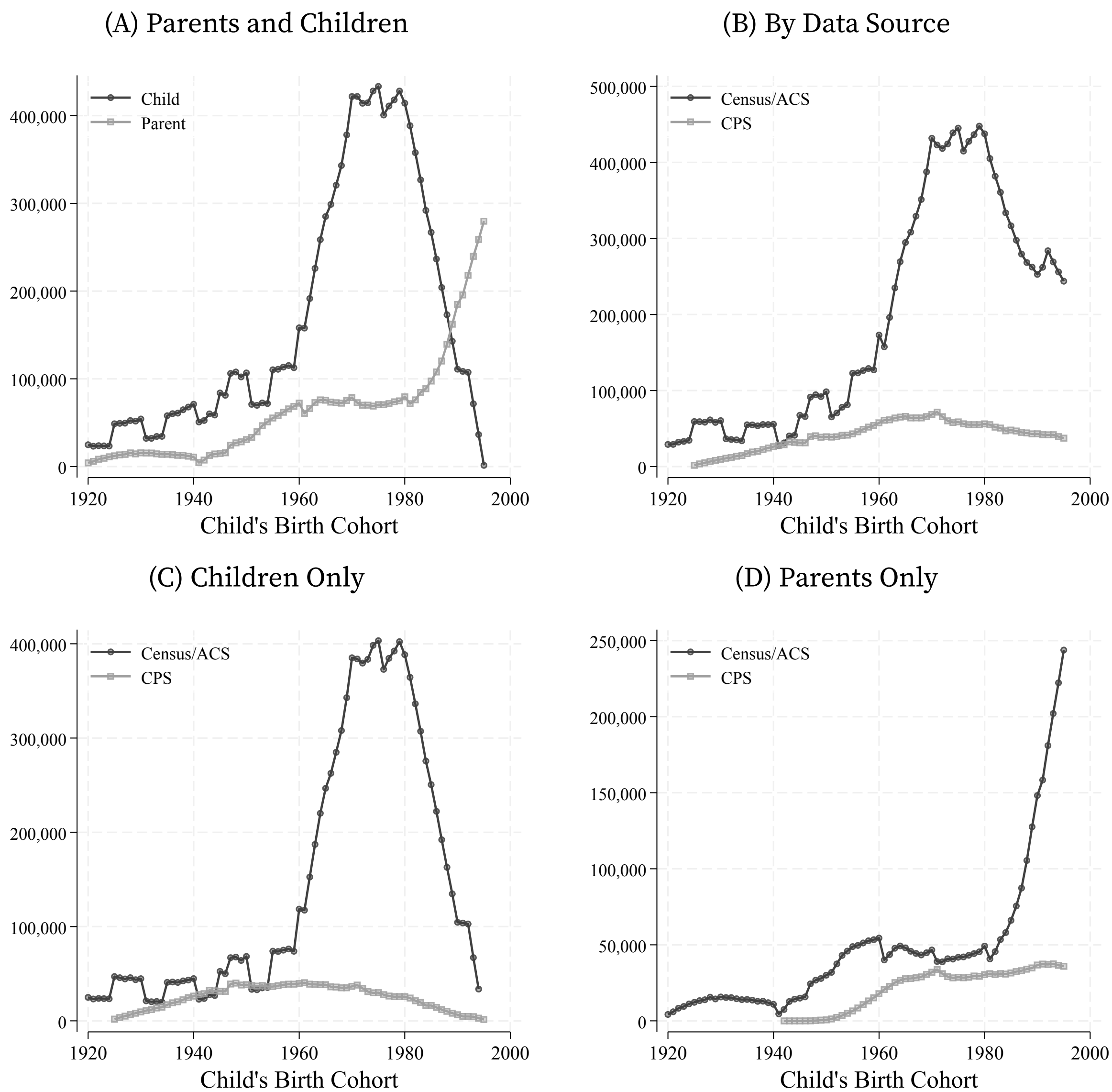
Appendix D. Long-run mobility: additional results

Sample construction. Figure D.1 documents the number of observations used to construct the parent and child marginal income distributions for each cohort. The decennial Census supplies the early cohorts and the largest samples throughout the period in which it is available. The ASEC begins contributing annual observations in 1968, and the ACS replaces the Census long form after 2000. The combination permits annual cohort estimates even though no dataset follows a child and their own parent.

Because both generations are observed between ages 30 and 45, the surveys that contribute to the two marginal distributions differ mechanically across cohorts. For a given child birth cohort, the child distribution is drawn from surveys taken 30–45 years later. The parent distribution is drawn from survey years in which parents aged 30–45 are observed with a co-resident own child from that cohort. The early and late endpoints therefore have fewer contributing survey years, while the decennial spacing of the Census produces visible variation in the number of parent observations.

Figure D.2A shows our main results are not driven by income coverage or sampling frame for any single data source, as they all provide a consistent picture of mobility.

FIGURE D.1. Sample Sizes Used to Construct Long-run US Mobility



Notes: The figure reports the number of observations with nonmissing family income used to construct each child's birth-cohort distribution. Panel A separates observations in the parent and child marginal distributions. Panel B reports total observations by source, and Panels C and D report the same source decomposition separately for children and parents. The sources are the decennial Census, the ACS, and the CPS ASEC. Sample restrictions are described in Section 4.1.

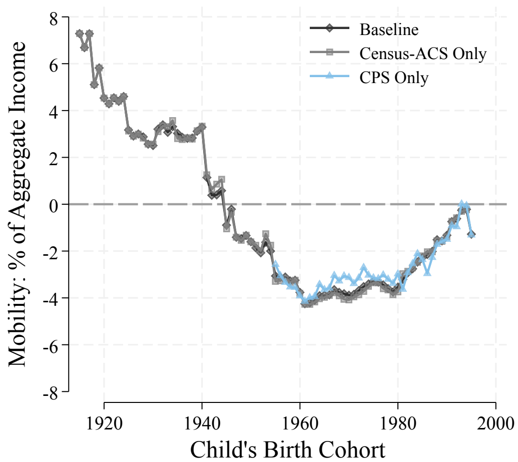
Age window. Our baseline results record income for children and parents when they are between 30 and 45 years of age. Figure D.2B shows how our results change with different age windows. The results broadly agree. Moving to younger ages (30–35 years of age) increases mobility, but the results are more similar for ages 35–40 and 40–45.

Alternative income measures. Our baseline results use total family income because the composition of families and the allocation of market work within them changed substantially over the sample period. Figure D.2C instead constructs both marginal distributions from the wage and salary income of the focal individual. The individual-income series has the same broad U-shaped pattern as the baseline, but declines earlier and reaches a lower mid-century trough. The long-run decline and subsequent recovery in net mobility are therefore not an artifact of pooling income within families, although their magnitude depends on the income concept.

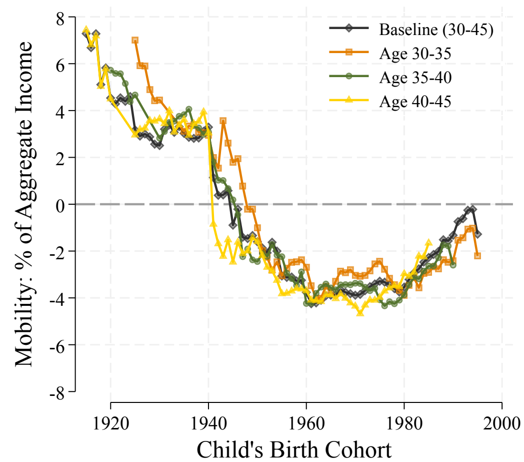
Robustness on results for Black and White Americans. Figure D.3 provides three complementary views of the results on the mobility of Black and White Americans. Within-group Lorenz mobility evaluates each individual relative to the income distribution of their own racial group, whereas the main analysis evaluates all individuals relative to the population distribution. The much smaller within-group series for Black Americans shows that the positive horizontal component in Figure 8B primarily captures movement relative to other groups. The rank series makes this distinction particularly clear: the large positional gains for Black Americans translate into smaller gains in aggregate-income shares once the distribution is used to price those movements. The family- and individual-income comparisons preserve the principal racial trajectories, although individual income produces larger mid-century declines, particularly for White Americans.

FIGURE D.2. Long-run Net Mobility: Robustness

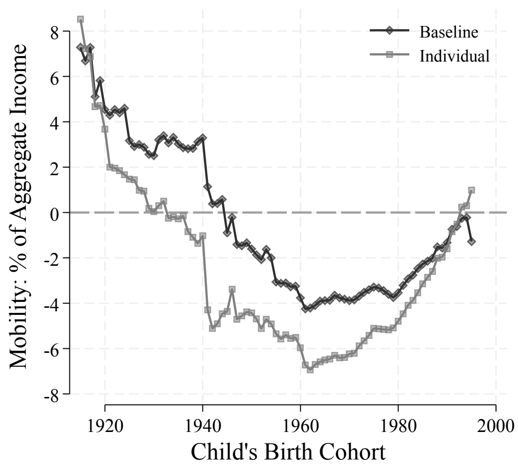
(A) Data Source



(B) Sample Age

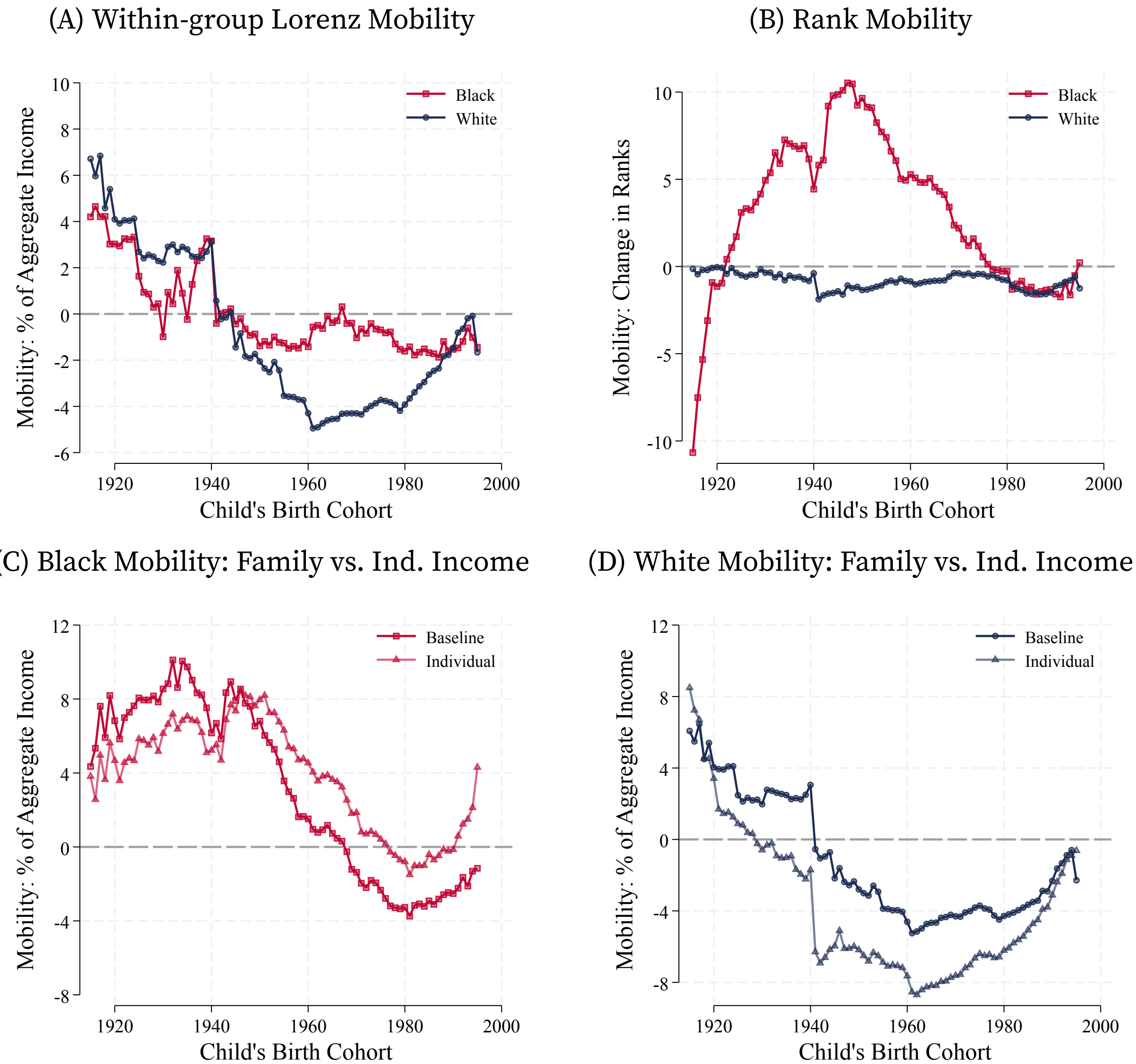


(C) Family vs. Individual Income



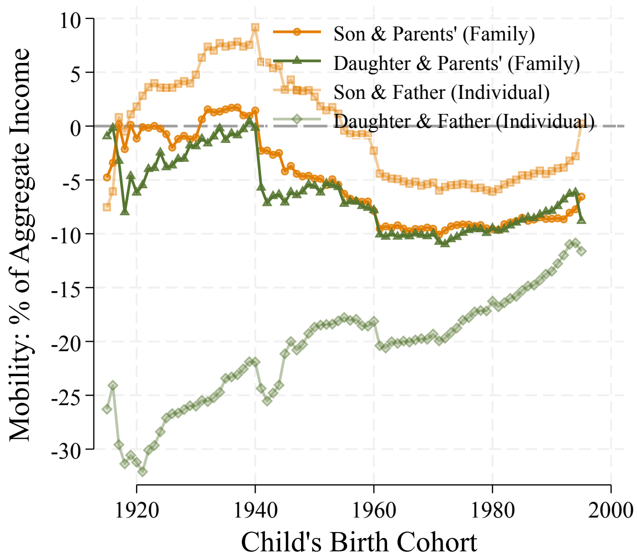
Notes: The baseline series reports average signed mobility constructed from total family income using all available data and an age window for income between 30 and 45 years of age. Panel A contrasts with different data sources that are combined for the baseline results. Panel B contrasts with different sample ages. Panel C contrasts with the individual series that replaces family income with the focal person's wage and salary income in both generations. The horizontal dashed line denotes zero net mobility.

FIGURE D.3. Long-run Net Mobility by Race: Additional Results



Notes: Panel A constructs Lorenz ordinates from the income distribution within each race group. Panel B reports the average change in population income rank, measured in percentile-rank points. Panels C and D compare population-Lorenz mobility constructed from family income and focal-individual wage and salary income for Black and White Americans, respectively. Except for rank mobility, all outcomes are percentage points of aggregate income.

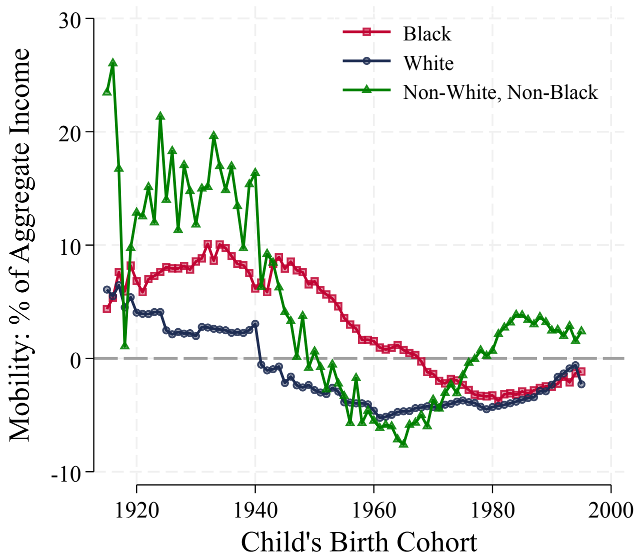
FIGURE D.4. Long-run Net Mobility: Sons vs. Daughters



Notes: The family-income series compare sons or daughters with the family income of parents observed with a child of the corresponding sex. The individual-income series compare sons' and daughters' wage and salary income with that of fathers observed with a child of the corresponding sex. All series report average signed mobility in Lorenz ordinates by the child's birth cohort and are expressed in percentage points of aggregate income for the corresponding income concept.

Results for daughters and sons. Figure D.4 reports the corresponding comparisons separately for sons and daughters. With family income, the two series converge and are similar for the cohorts born after 1960. The individual-income comparison is different: father-daughter mobility is substantially more negative than father-son mobility, especially for the early cohorts. This divergence is consistent with the large change in women's market work over the period and supports the use of family income as the baseline long-run outcome.

FIGURE D.5. Long-run Net Mobility Including Non-White, Non-Black Americans

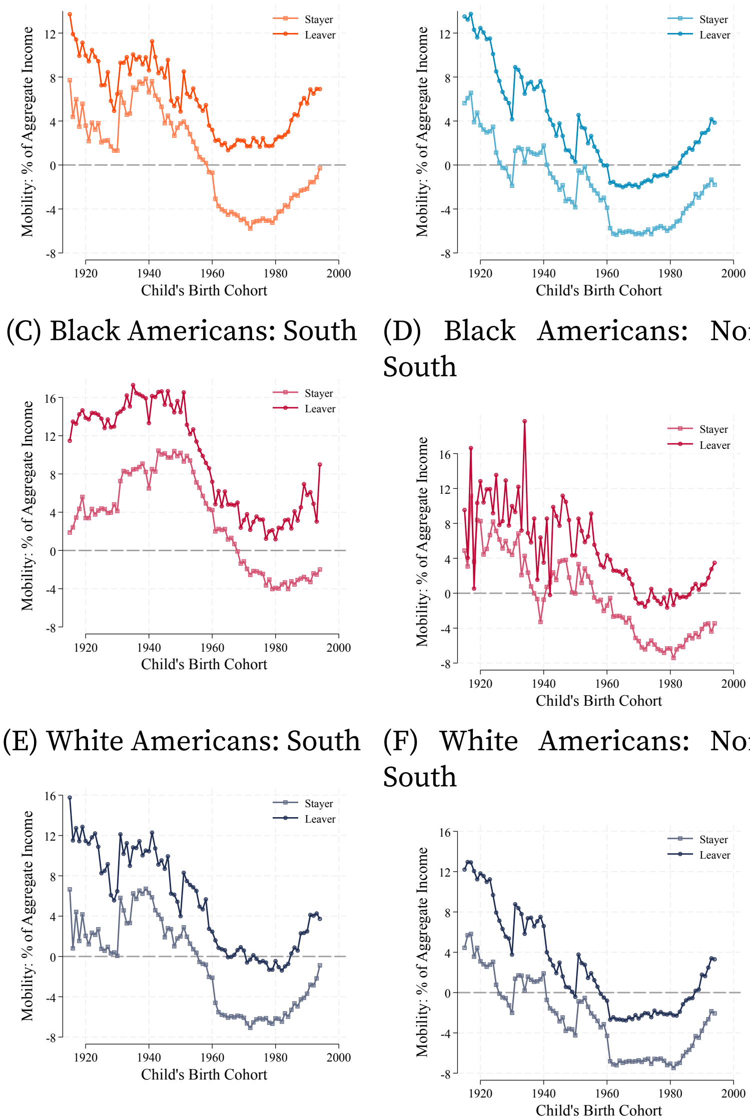


Notes: The figure reports average signed mobility in population Lorenz ordinates for Black, White, and non-White, non-Black Americans. The last category combines the remaining harmonized race groups so that its definition can be maintained across the Census, ACS, and ASEC. The outcome is expressed in percentage points of aggregate family income. The horizontal dashed line denotes zero net mobility.

Alternative group definitions. Figure D.5 adds the harmonized non-White, non-Black category omitted from the main analysis. Its estimates are volatile before 1955 because the relevant cells are small. Moreover, the Hart–Celler Act and the subsequent change in immigration altered the composition of this residual group after 1965, making comparisons across cohorts difficult to interpret. We therefore retain the Black–White comparison as the main racial result.

Migration and mobility. Figure D.6 examines heterogeneity by adult migration. Among children born either inside and outside the South, those living outside their birth region at ages 30–45 generally have greater net mobility than those who remain. The difference is visible for both racial groups and is especially persistent among Black Americans born in the South. These estimates are descriptive: migration may respond to the same opportunities that shape adult income, and the repeated cross-sections do not link a migrant to their own parent.

FIGURE D.6. Mobility of Stayers and Leavers



(A) South

(B) Non-South

(C) Black Americans: South

(D) Black Americans: Non-South

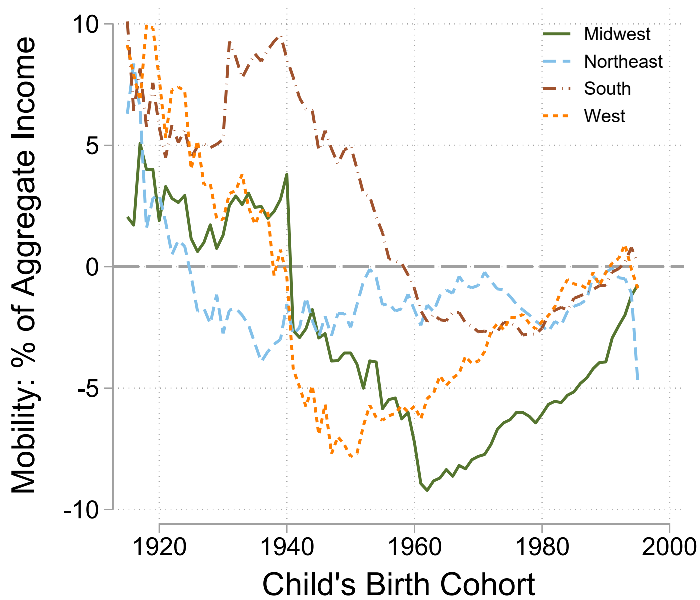
(E) White Americans: South

(F) White Americans: Non-South

Notes: The figure uses Census observations and classifies children by Census region of birth. A stayer lives in the same Census region at ages 30–45, while a leaver lives in a different region. Panels A and B report the population comparisons; the remaining panels divide them by race. Within each birthplace-region, cohort, and race comparison, the same regional parent distribution is used for stayers and leavers. All series report average signed mobility in Lorenz ordinates and are expressed in percentage points of aggregate family income.

Alternative region definitions. Figure D.7 disaggregates our main estimate of the differences in mobility by region (Figure 9A). Breaking the non-South region into its constituent census regions shows that the differences in the timing and extent of the

Figure D.7. Differences by Census Region



Notes: The figure reports average signed mobility in Lorenz ordinates by the child's region of birth using the four major census areas. It corresponds to Panel A in the main text which compares children born in the Census South region with children born elsewhere. Parents are classified by adult residence, while children are classified by birthplace.

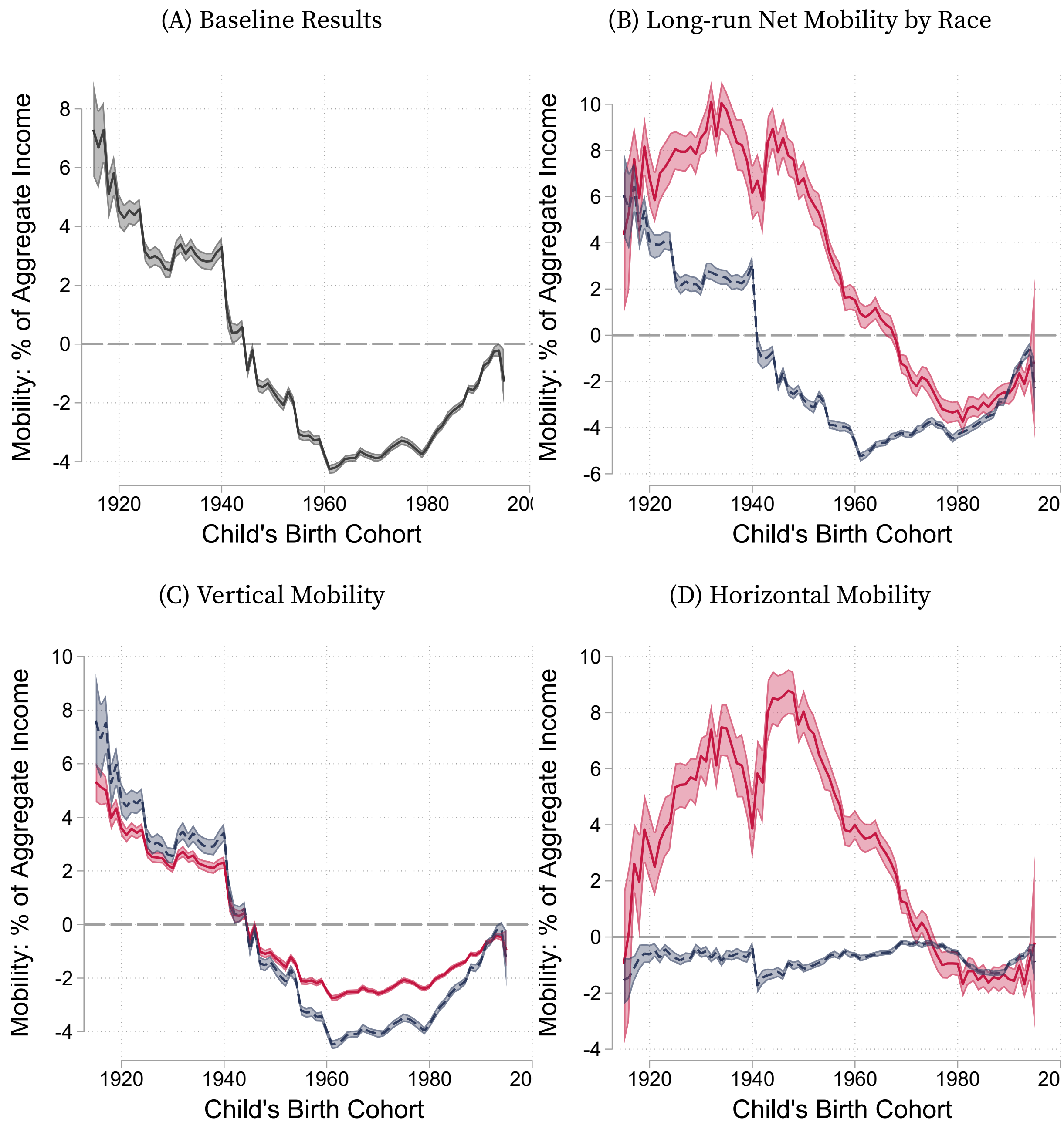
decline are non-trivial. For example the West recovers earlier than the Midwest and the Northeast has a smaller decline in mobility, but its decline occurs earlier. Nevertheless, the overall pattern differs sharply for the South vs non-South regions supporting our choice to aggregate.

D.1. Confidence intervals for mobility estimates

We construct confidence intervals using a stratified bootstrap procedure. We compute 200 replications to construct 95% confidence intervals. We stratify by year and underlying data source, which enforces identical composition between the Census-ACS and CPS samples in each year. We resample at the household level within strata which are then used to construct child–parent birth cohorts.

Overall, our confidence intervals are narrow, enabling us to reject the null hypothesis of stable mobility. Moreover, despite wider confidence intervals for Black Americans, reflecting racial composition over time, we find differences across racial groups are statistically significant.

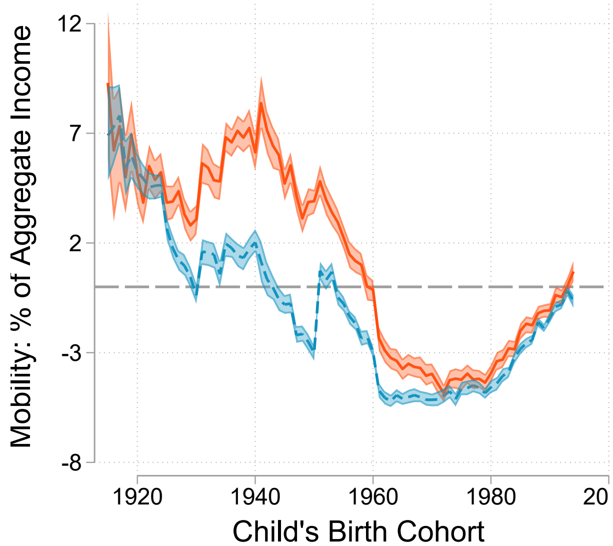
FIGURE D.8. Long-run Net Mobility



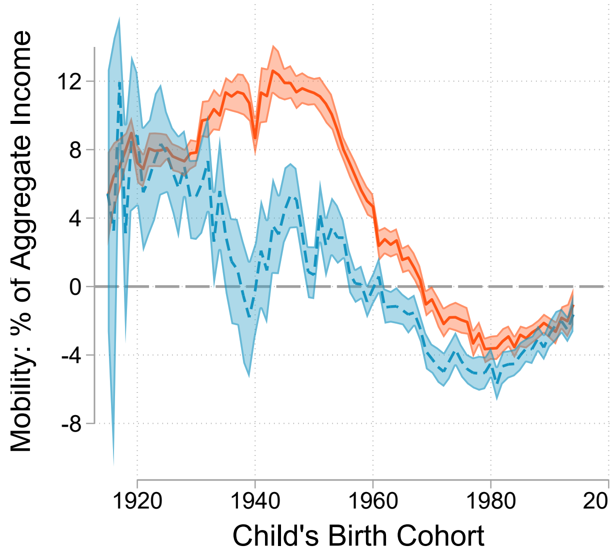
Notes: Panel A reports average signed mobility in Lorenz ordinates, as defined in equation (11), by the child's birth cohort. The series is the difference between the average Lorenz ordinate of the child and parent marginal family-income distributions and is expressed in percentage points of aggregate income. Panel B reports average signed mobility in Lorenz ordinates separately for Black and White Americans, by the child's birth cohort. Group mobility is the average change in aggregate-income shares for members of that group, as characterized in Lemma 2. Panels A and B decompose the group-specific net mobility in Panel B into the vertical and horizontal components defined in equation (3).

FIGURE D.9. The Geography of Mobility

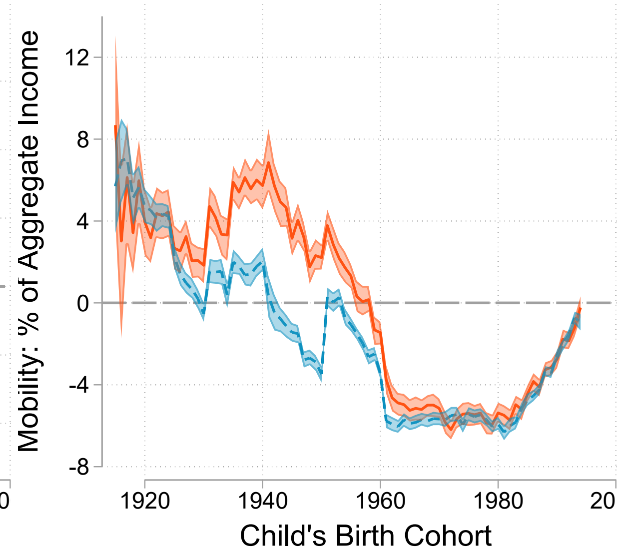
(A) South vs Non-South



(B) Mobility of Black Americans by Region



(C) Mobility of White Americans by Region



Notes: The figure reports average signed mobility in Lorenz ordinates by the child's region of birth. Panel A compares children born in the Census South region with children born elsewhere. Panels B and C make the same comparison separately for Black and White Americans. Parents are classified by adult residence, while children are classified by birthplace.